AEROFIT

Aerofit is a leading brand in the field of fitness equipment. Aerofit provides a product range including machines such as treadmills, exercise bikes, gym equipment, and fitness accessories to cater to the needs of all categories of people.

```
#Importing libraries -
import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns
```

Problem Statement:

The market research team at AeroFit wants to identify the characteristics of the target audience for each type of treadmill offered by the company, to provide a better recommendation of the treadmills to the new customers. The team decides to investigate whether there are differences across the product with respect to customer characteristics.

Perform descriptive analytics to create a customer profile for each AeroFit treadmill product by developing appropriate tables and charts. For each AeroFit treadmill product, construct two-way contingency tables and compute all conditional and marginal probabilities along with their insights/impact on the business.

Analysing Basic Metrics:

The Aerofit dataset case study aims to analyze and derive meaningful insights from a comprehensive dataset related to fitness and exercise. The scope of this study encompasses various facets of customer experience and to conduct a thorough dataset exploration to identify patterns, correlations, and outliers.

Product Purchased: KP281, KP481, or KP781

Age: In years

Gender: Male/Female

Education: In years

MaritalStatus: Single or partnered

Usage: The average number of times the customer plans to use the treadmill each week.

Income: Annual income (in $)

Fitness: Self-rated fitness on a 1-to-5 scale, where 1 is the poor shape and 5 is the excellent shape.

Miles: The average number of miles the customer expects to walk/run each week

Product Portfolio:

The KP281 is an entry-level treadmill that sells for $1,500.

The KP481 is for mid-level runners that sell for $1,750.

The KP781 treadmill is having advanced features that sell for $2,500.

1. Import The Dataset And Do Usual Data Analysis Steps Like Checking The Structure & Characteristics Of The Dataset.

a. The data type of all columns in the “customers” table. Hint: We want you to display the data type of each column present in the dataset. b. You can find the number of rows and columns given in the dataset Hint: We want you to find the shape of the dataset. ○ Check for the missing values and find the number of missing values in each column

Basic Analysis Of The Data - EDA

Importing the dataset

```python
data= pd.read_csv('aerofit_treadmill.csv')
data.head()
```

	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
0	KP281	18	Male	14	Single	3	4	29562	112
1	KP281	19	Male	15	Single	2	3	31836	75
2	KP281	19	Female	14	Partnered	4	3	30699	66
3	KP281	19	Male	12	Single	3	3	32973	85
4	KP281	20	Male	13	Partnered	4	2	35247	47

Getting Basic Info About The Dataset

```python
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   Product        180 non-null    object
 1   Age            180 non-null    int64 
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64 
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64 
 6   Fitness        180 non-null    int64 
 7   Income         180 non-null    int64 
 8   Miles          180 non-null    int64 
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

*>**Insight:** The dataset consists of 3 columns containing object datatype, while 6 column contains integer datatype*

Shape Of Dataset

```python
data.shape
```

```
(180, 9)
```

*>**Insight:** There are 180 rows and 9 columns in the dataset*

Duplicate Values

```python
data.duplicated().any()
```

```
False
```

*>**Insight:** There are no duplicate rows in dataset.*

Missing Values

```python
data.isna().any()
```

```
Product          False
Age              False
Gender           False
Education        False
MaritalStatus    False
Usage            False
Fitness          False
Income           False
Miles            False
dtype: bool
```

*>**Insight:** There are no missing values in dataset.*

Statistical Summary

```python
data.describe()
```

	Age	Education	Usage	Fitness	Income	Miles
count	180.000000	180.000000	180.000000	180.000000	180.000000	180.000000
mean	28.788889	15.572222	3.455556	3.311111	53719.577778	103.194444
std	6.943498	1.617055	1.084797	0.958869	16506.684226	51.863605
min	18.000000	12.000000	2.000000	1.000000	29562.000000	21.000000
25%	24.000000	14.000000	3.000000	3.000000	44058.750000	66.000000
50%	26.000000	16.000000	3.000000	3.000000	50596.500000	94.000000
75%	33.000000	16.000000	4.000000	4.000000	58668.000000	114.750000
max	50.000000	21.000000	7.000000	5.000000	104581.000000	360.000000

*>**Insight:** This function calculates summary statistics, including count, mean, standard deviation, minimum, 25th percentile, median (50th percentile), 75th percentile, and maximum, for numerical columns within the DataFrame. These statistics aid in comprehending the central tendency, spread, and distribution of the data.*

```
1 data.describe(include= 'object').T
```

	count	unique	top	freq
Product	180	3	KP281	80
Gender	180	2	Male	104
MaritalStatus	180	2	Partnered	107

*>**Insight:** This command retrieves information about categorical data within the DataFrame, including the count of observations, the number of unique categories, the most frequent category, and its frequency for each categorical column.*

Non-Graphical & Graphical Analysis: Value counts and unique attributes

Univariate Analysis

Unique Values

```
1 data.nunique()
```

```
Product           3
Age              32
Gender            2
Education         8
MaritalStatus     2
Usage             6
Fitness           5
Income           62
Miles            37
dtype: int64
```

*>**Insight**:* Product: 3 unique treadmill products are offered by Aerofit. Age: There are 32 different age groups represented among the customers. Gender: Customers are categorized into 2 genders: Male and Female. Education: Customers have completed education levels ranging from 12 to 21 years, with 8 unique levels. MaritalStatus: Customers are classified as either Single or Partnered. Usage: Customers plan to use the treadmill varying from 2 to 7 times a week, with 6 different frequencies. Fitness: Customers rate their fitness levels on a scale of 1 to 5, with 5 unique ratings. Income: There are 62 different income levels represented among the customers. Miles: Customers expect to walk/run varying distances from 38 to 360 miles per week, with 37 unique distances.

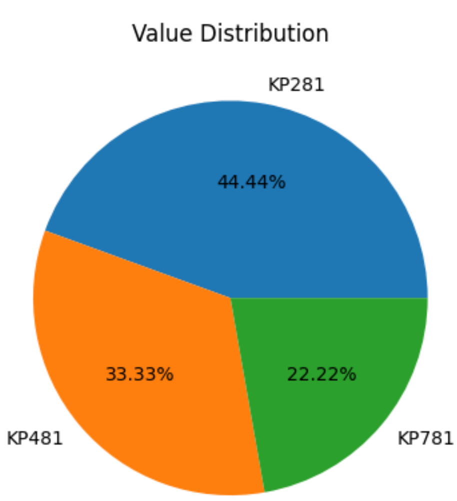
Value Counts Of Categorical Data

Product

```
1 data['Product'].value_counts()
```

```
KP281    80
KP481    60
KP781    40
Name: Product, dtype: int64
```

```
1 value= data['Product'].value_counts()
2 type_data= data['Product'].value_counts().index
3 plt.pie(value, labels= type_data, autopct= '%2.2f%%')
4 plt.title('Value Distribution')
5 plt.show()
```



Insight And Recommendation

*>**Insight:** The distribution of purchases among Aerofit's treadmill products shows that the KP281 model has the highest frequency of transactions (80), followed by KP481 (60), and KP781 (40), indicating varying levels of popularity and demand among customers. The popularity of the KP281 treadmill suggests that customers might prioritize affordability and basic features over advanced functionalities offered by higher-end models, reflecting a segment of the market seeking cost-effective fitness solutions.*

*>**Recommendation:** Given the high demand for the KP281 model, Aerofit should ensure ample availability of this product in stores and online channels, while also considering potential expansion of its product line to include more entry-level options to cater to budget-conscious customers.*

To capitalize on the popularity of the KP481 model, Aerofit should focus on highlighting its unique features and advantages compared to other mid-level treadmills in the market, such as durability, performance, and advanced technology, through targeted marketing campaigns and customer engagement initiatives.

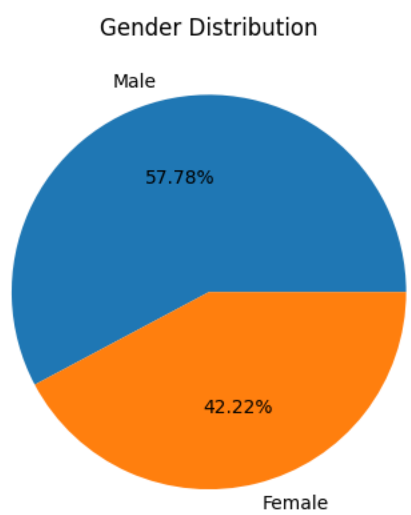
While the KP781 treadmill has a lower frequency of purchases, its status as a premium product indicates a niche market segment interested in high-end fitness equipment. Aerofit should focus on enhancing the perceived value of the KP781 model by emphasizing its cutting-edge features, customization options, and premium build quality, targeting affluent customers willing to invest in top-tier fitness solutions. Additionally, offering personalized consultations and demonstrations could help showcase the unique benefits of the KP781 treadmill and attract discerning buyers.

Gender

```
data['Gender'].value_counts()
```

```
Male      104
Female     76
Name: Gender, dtype: int64
```

```
value= data['Gender'].value_counts()
type_data= data['Gender'].value_counts().index
plt.pie(value, labels= type_data, autopct= '%2.2f%%')
plt.title('Gender Distribution')
plt.show()
```



Insight And Recommendation

*>**Insight:** The dataset indicates that there are 104 entries for male customers and 76 entries for female customers, suggesting a higher representation of male customers in the dataset compared to female customers. This gender imbalance may reflect broader trends in the fitness equipment industry, where males traditionally comprise a larger portion of the customer base.*

*>**Recommendation:** Develop gender-inclusive marketing strategies that resonate with both male and female audiences. Avoid gender stereotypes and ensure that advertising and promotional materials appeal to diverse gender identities and preferences.*

Consider diversifying product features to cater to the preferences and needs of both male and female customers. Conduct market research to identify specific features or functionalities that appeal to different gender demographics and incorporate these insights into product development processes.

Implement targeted outreach efforts to engage with underrepresented gender groups. This may involve collaborating with fitness influencers, hosting events or workshops specifically tailored to female customers, and leveraging social media platforms to connect with diverse audiences.

Establish a feedback mechanism to collect insights from male and female customers regarding their experiences with Aerofit products and services. Use this feedback to continuously refine offerings and ensure that they meet the evolving needs of all customer segments.

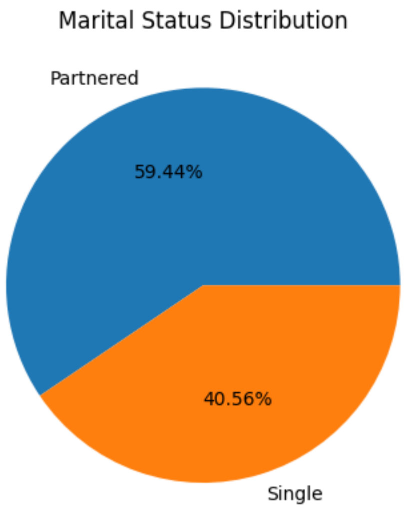
Foster an inclusive fitness community that welcomes individuals of all genders and encourages participation regardless of gender identity. This may involve organizing gender-neutral fitness classes or creating online forums where customers can share their fitness journeys and support one another.

Marital Status

```
data['MaritalStatus'].value_counts()
```

```
Partnered    107
Single        73
Name: MaritalStatus, dtype: int64
```

```
value= data['MaritalStatus'].value_counts()
type_data= data['MaritalStatus'].value_counts().index
plt.pie(value, labels= type_data, autopct= '%2.2f%%')
plt.title('Marital Status Distribution')
plt.show()
```



Insight And Recommendation

*>**Insight:** The dataset indicates that there are 107 entries for partnered individuals and 73 entries for single individuals, revealing a higher representation of partnered customers compared to single customers. This distribution reflects the diversity of marital status among Aerofit's clientele, with implications for marketing strategies and product offerings.*

*>**Recommendation:** Develop targeted marketing campaigns that cater to the specific needs and preferences of both partnered and single individuals. Tailor messaging and promotional materials to resonate with each demographic, highlighting how Aerofit's products can enhance their fitness journeys within their respective lifestyles.*

Offer product customization options to accommodate the diverse requirements of partnered and single customers. For example, consider providing flexible payment plans or bundle deals for partnered customers, while offering solo-focused fitness programs or accessories for single individuals.

Foster a sense of community among Aerofit customers by organizing events, workshops, or online forums that cater to the interests of both partnered and single individuals. Encourage social interaction and networking opportunities to create a supportive environment for all members, regardless of marital status.

Implement mechanisms for collecting feedback from partnered and single customers to understand their unique needs and challenges. Use this feedback to refine existing products and services, as well as to develop new offerings that address specific pain points or aspirations within each demographic group.

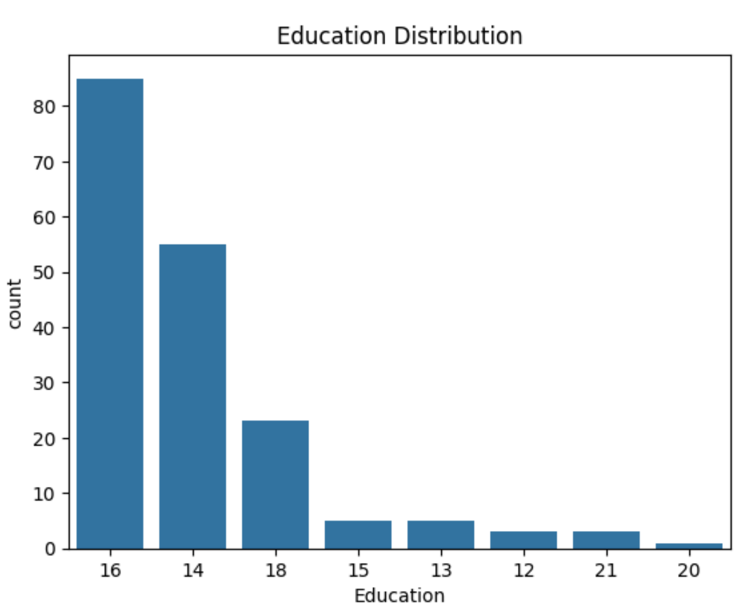
Ensure that Aerofit's brand messaging and imagery are inclusive and representative of diverse marital status backgrounds. Avoid stereotypes and biases in advertising materials, and strive to portray a broad spectrum of customer experiences and lifestyles.

Education

```
data['Education'].value_counts()
```

```
16    85
14    55
18    23
15     5
13     5
12     3
21     3
20     1
Name: Education, dtype: int64
```

```
plt.title('Education Distribution')
sns.countplot(data= data, x='Education', order= data['Education'].value_counts().index)
plt.show()
```



Insight And Recommendation

*>**Insight:**The education levels of customers vary, with the majority having completed 16 years of education, followed by 14 and 18 years. There are also smaller numbers of customers with 15, 13, 12, 21, and 20 years of education.*

*>**Recommendation:** Tailored Product Information: Provide educational resources and product information that cater to customers across different education levels. Ensure that marketing materials are accessible and easy to understand for individuals with varying levels of education.*

Utilize a variety of communication channels to reach customers with different education levels. While some may prefer written materials, others may respond better to visual or interactive content. Adapt communication strategies accordingly to ensure maximum engagement.

Offer training and support programs that accommodate the learning needs of customers with different education levels. Provide clear instructions, demonstrations, and ongoing assistance to help customers make the most of their fitness equipment and achieve their fitness goals effectively.

Implement a feedback mechanism to gather insights from customers regarding their educational backgrounds and preferences. Use this information to tailor future product development, marketing campaigns, and customer support initiatives to better serve the diverse educational needs of the customer base.

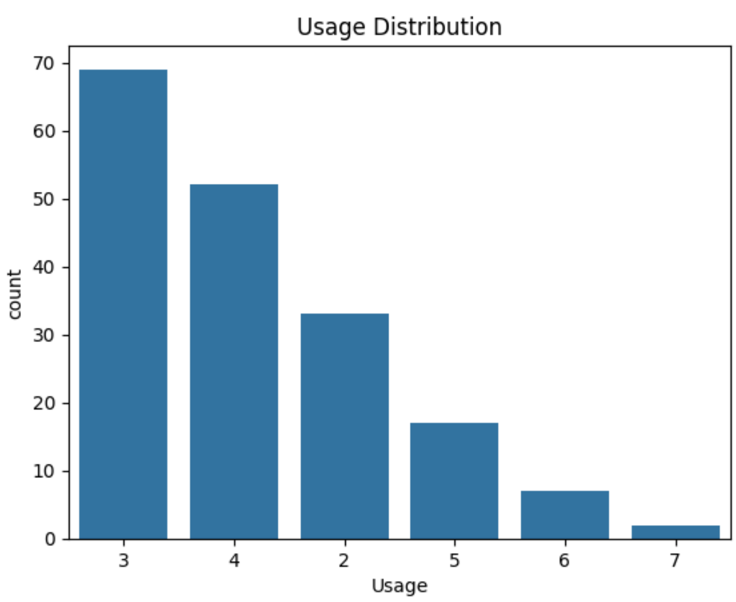
Explore partnerships with educational institutions or fitness certification programs to offer specialized training and educational opportunities for customers interested in expanding their knowledge and skills in fitness and exercise science.

Usage

```
data['Usage'].value_counts()
```

```
3    69
4    52
2    33
5    17
6     7
7     2
Name: Usage, dtype: int64
```

```
plt.title('Usage Distribution')
sns.countplot(data= data, x='Usage', order= data['Usage'].value_counts().index)
plt.show()
```



Insight And Recommendation

*>**Insight:**The dataset indicates that customers have different usage patterns for the treadmills they purchased, with the majority planning to use them 3 times a week, followed by 4 times a week. Smaller groups of customers plan to use the treadmill 2, 5, 6, or 7 times a week, respectively.*

*>**Recommendation:** Customized Training Plans: Provide customers with customized training plans tailored to their intended usage frequency. Offer resources and guidance on how to maximize workout efficiency based on the number of sessions per week planned by the customer.*

Implement a maintenance reminder system to encourage regular upkeep of the treadmills, especially for customers planning to use them frequently. Provide tips on proper maintenance practices to ensure optimal performance and longevity of the equipment.

Develop tools or apps that allow customers to track their treadmill usage and progress towards their fitness goals. Offer features such as workout logs, performance metrics, and goal-setting functionalities to enhance the overall user experience.

Create promotional offers or incentives targeting customers with specific usage frequencies. For example, offer discounts on accessories or extended warranties for customers planning to use the treadmill more frequently as a way to incentivize their commitment to regular exercise.

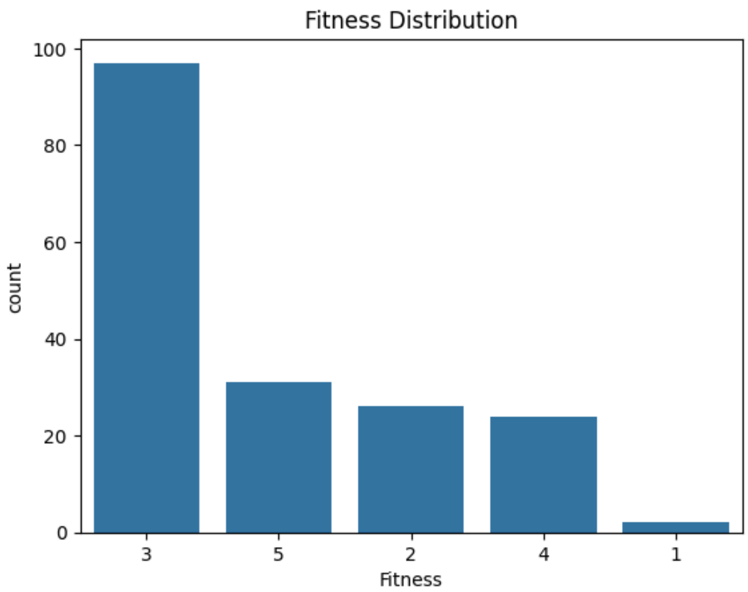
Provide comprehensive customer support resources, including online tutorials, troubleshooting guides, and access to knowledgeable support staff, to assist customers in optimizing their treadmill usage experience and addressing any issues or concerns they may encounter.

Fitness

```
data['Fitness'].value_counts()
```

```
3    97
5    31
2    26
4    24
1     2
Name: Fitness, dtype: int64
```

```
plt.title('Fitness Distribution')
sns.countplot(data= data, x='Fitness', order= data['Fitness'].value_counts().index)
plt.show()
```



Insight And Recommendation

*>**Insight:**The data reveals that customers rate their fitness levels on a scale of 1 to 5, with the majority rating themselves at level 3, followed by level 5. There are also significant numbers of customers rating themselves at levels 2, 4, and a smaller number at level 1.*

*>**Recommendation:** Customized Workout Plans: Offer customized workout plans and fitness programs tailored to customers' self-rated fitness levels. Provide resources and guidance that align with their current fitness capabilities and goals to ensure effective and safe workouts.*

Develop tools and features within fitness apps or equipment interfaces that allow customers to track their progress based on their self-rated fitness levels. Include features such as goal setting, performance tracking, and milestone achievements to motivate customers and enhance engagement.

Provide educational resources and content that cater to customers across different fitness levels. Offer tips, tutorials, and instructional videos designed to help customers improve their fitness level gradually and sustainably.

Utilize customer data and self-reported fitness levels to personalize product recommendations and promotions. Offer targeted suggestions for equipment upgrades, accessories, or training programs that align with customers' fitness goals and current abilities.

Foster a supportive online community where customers can share their fitness journeys, exchange tips, and support one another based on their self-reported fitness levels. Encourage participation in group challenges, virtual workouts, and forums to create a sense of camaraderie and accountability.

2. Detect Outliers

a. Find the outliers for every continuous variable in the dataset

Hint: We want you to use boxplots to find the outliers in the given dataset

b. Remove/clip the data between the 5 percentile and 95 percentile

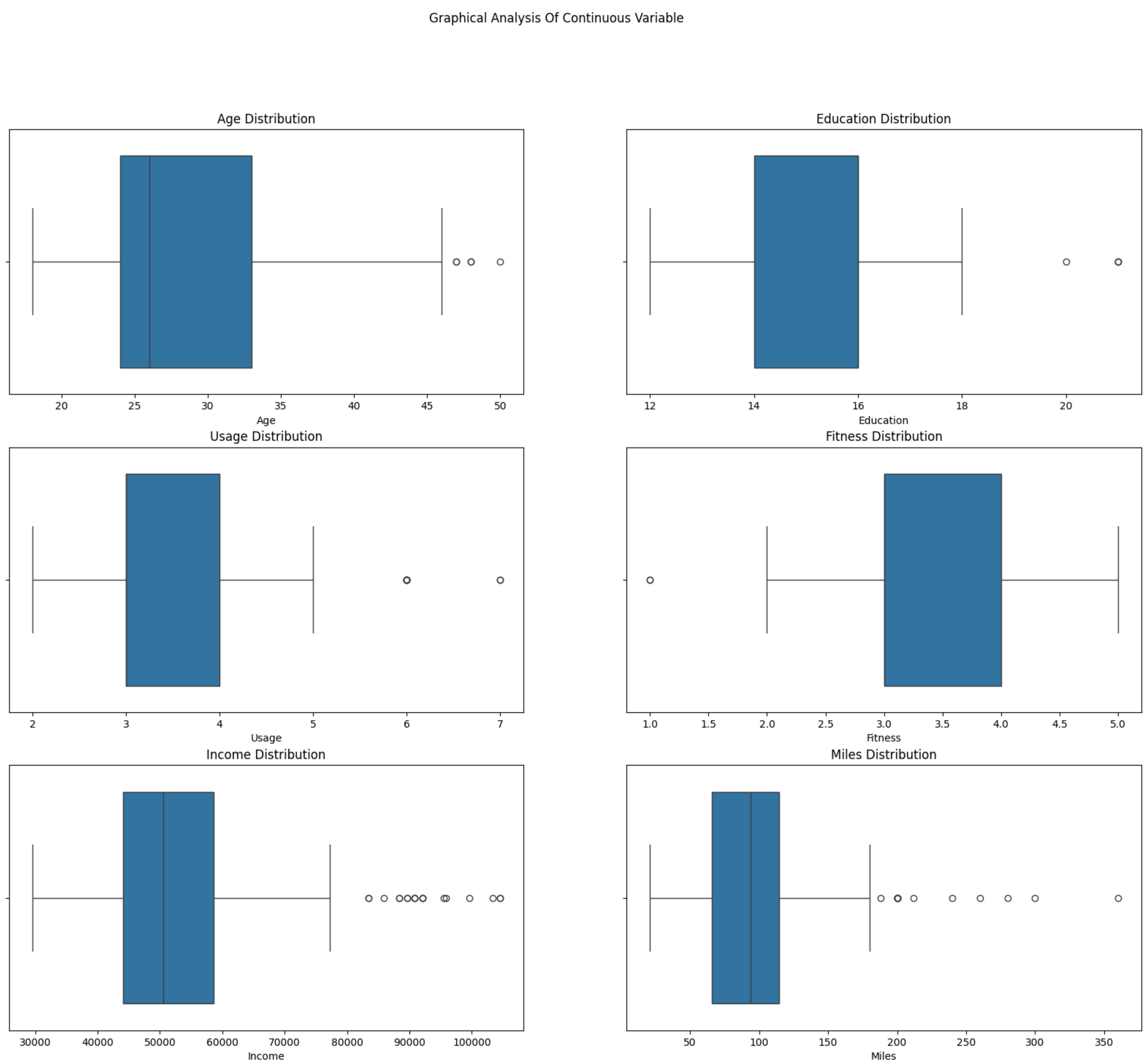
Hint: We want You to use np.clip() for clipping the data

```
data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   Product        180 non-null    object
 1   Age            180 non-null    int64 
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64 
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64 
 6   Fitness        180 non-null    int64 
 7   Income         180 non-null    int64 
 8   Miles          180 non-null    int64 
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

Finding Outliers

```
plt.figure(figsize= (20, 16))
plt.suptitle('Graphical Analysis Of Continuous Variable')

plt.subplot(3, 2, 1)
sns.boxplot(data= data, x='Age')
plt.title('Age Distribution')

plt.subplot(3, 2, 2)
sns.boxplot(data= data, x='Education')
plt.title('Education Distribution')

plt.subplot(3, 2, 3)
sns.boxplot(data= data, x='Usage')
plt.title('Usage Distribution')

plt.subplot(3, 2, 4)
sns.boxplot(data= data, x='Fitness')
plt.title('Fitness Distribution')

plt.subplot(3, 2, 5)
sns.boxplot(data= data, x='Income')
plt.title('Income Distribution')

plt.subplot(3, 2, 6)
sns.boxplot(data= data, x='Miles')
plt.title('Miles Distribution')

plt.show()
```



Treatement Of Outliers

Clipping The Data Between 5 Percentile and 95 Percentile

NOTE: *Certainly, we acknowledge the presence of outliers in our dataset. However, removing them entirely can result in a loss of valuable information, especially given the relatively small size of our dataset, which consists of only 180 rows. Therefore, instead of outright removal, we'll opt for a clipped approach, wherein the data will be adjusted to fall within a certain range (between the 5th and 95th percentiles).*

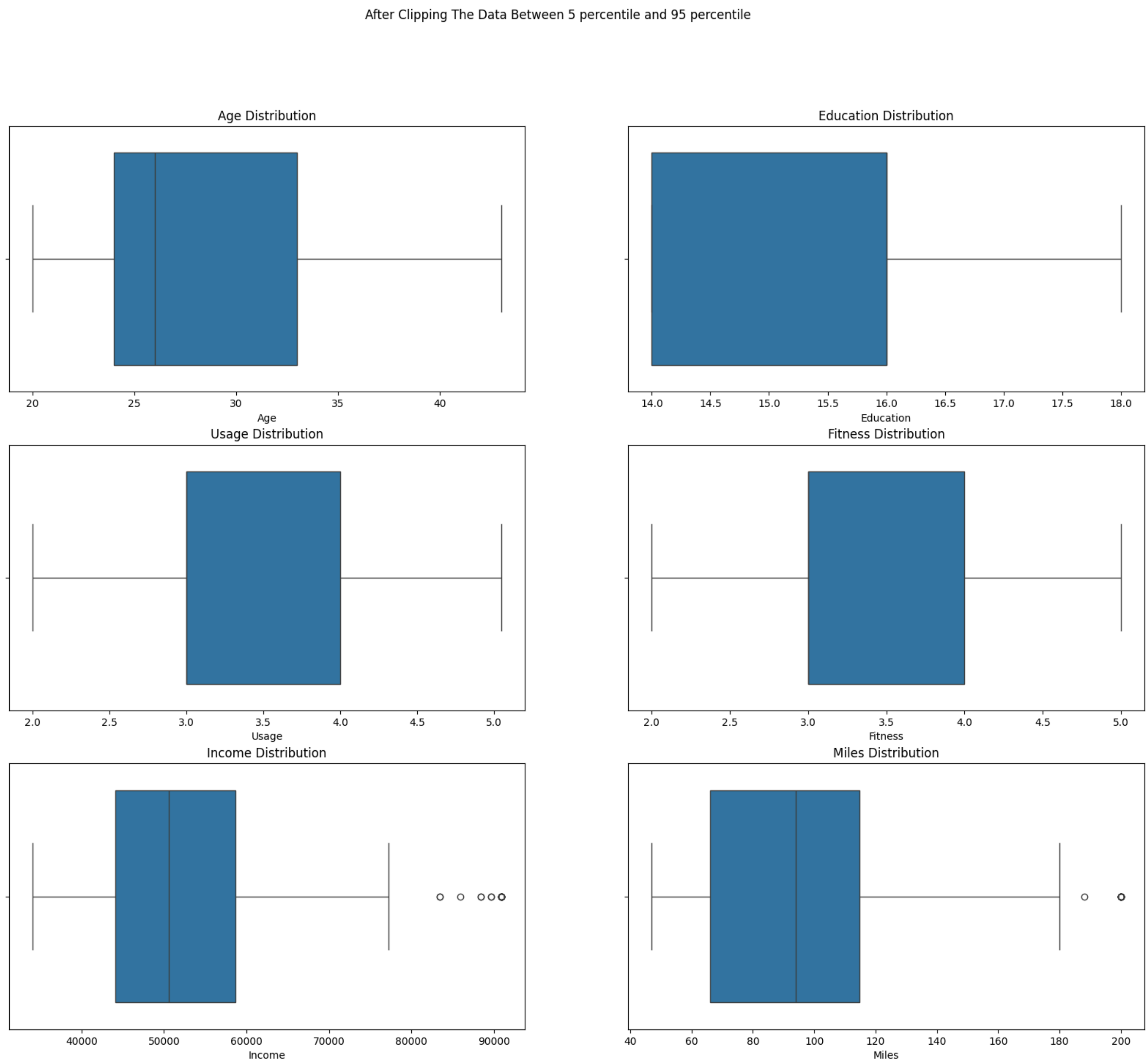
Please note that although the code provided demonstrates outlier treatment by clipping the data, this adjustment has not yet been incorporated into the subsequent analysis.

```
data_copy= data.copy()

Outliers_col=['Age','Education','Usage','Fitness', 'Income', 'Miles']
for col in Outliers_col:
    p5 = np.percentile(data_copy[col], 5)
    p95 = np.percentile(data_copy[col], 95)
    data_copy[col] = np.clip(data_copy[col], p5, p95)
```

After Treatement Of Outliers

After Clipping The Data Between 5 percentile and 95 percentile

```
plt.figure(figsize= (20, 16))
plt.suptitle('After Clipping The Data Between 5 percentile and 95 percentile')

plt.subplot(3, 2, 1)
sns.boxplot(data= data_copy, x='Age')
plt.title('Age Distribution')

plt.subplot(3, 2, 2)
sns.boxplot(data= data_copy, x='Education')
plt.title('Education Distribution')

plt.subplot(3, 2, 3)
sns.boxplot(data= data_copy, x='Usage')
plt.title('Usage Distribution')

plt.subplot(3, 2, 4)
sns.boxplot(data= data_copy, x='Fitness')
plt.title('Fitness Distribution')

plt.subplot(3, 2, 5)
sns.boxplot(data= data_copy, x='Income')
plt.title('Income Distribution')

plt.subplot(3, 2, 6)
sns.boxplot(data= data_copy, x='Miles')
plt.title('Miles Distribution')

plt.show()
```



*>**Insight:**We've identified outliers within our dataset and have taken steps to address them. The data has been adjusted, ensuring that values now fall within a range defined by the 5th and 95th percentiles.*

3. Check If Features Like Marital Status, Gender And Age Have Any Effect On The Product Purchased

a. Find if there is any relationship between the categorical variables and the output variable in the data.

Hint: We want you to use the count plot to find the relationship between categorical variables and output variables.

b. Find if there is any relationship between the continuous variables and the output variable in the data.

Hint: We want you to use a scatter plot to find the relationship between continuous variables and output variables.

```
1 data.info()
```

```
<class 'pandas.core.frame.DataFrame'>
RangeIndex: 180 entries, 0 to 179
Data columns (total 9 columns):
 #   Column         Non-Null Count  Dtype 
---  ------         --------------  ----- 
 0   Product        180 non-null    object
 1   Age            180 non-null    int64 
 2   Gender         180 non-null    object
 3   Education      180 non-null    int64 
 4   MaritalStatus  180 non-null    object
 5   Usage          180 non-null    int64 
 6   Fitness        180 non-null    int64 
 7   Income         180 non-null    int64 
 8   Miles          180 non-null    int64 
dtypes: int64(6), object(3)
memory usage: 12.8+ KB
```

Converting Ages, Education, Fitness, Usage, Incomes and Miles To Category/Bins For Better Analysis.

```python
age_rng = [0,21,35,45,float("inf")]
bin_lbl = ["Teen","Adult","Middle Aged","Elder"]
data['Age_cat'] = pd.cut(data['Age'],bins = age_rng, labels = bin_lbl)

edu_bins = [0,13,18,float('inf')]
edu_lbl = ['Low Edu', 'Moderate Edu', 'High Edu']
data["Education_cat"] = pd.cut(data["Education"], bins = edu_bins, labels = edu_lbl)

bin_mils = [0,50,100,200,float('inf')]
mils_lbl = ["Light Activity", "Moderate Activity", "Active Lifestyle", "Fitness Enthusiast"]
data["Miles_cat"] = pd.cut(data["Miles"], bins = bin_mils, labels = mils_lbl)

bin_inc = [0,40000,60000,80000,float("inf")]
inc_lbl = ["Low", "Moderate", "High Income", "Very High"]
data["Income_cat"] = pd.cut(data["Income"], bins = bin_inc, labels = inc_lbl)

data['Fitness_cat']= data.Fitness
data["Fitness_cat"].replace({1:"Poor Shape", 2:"Bad Shape", 3:"Average Shape", 4:"Good Shape", 5:"Excellent Shape"}, inplace=True)

use_bins = [0,3,5,float('inf')]
use_lbl = ['Low Usage', 'Moderate Usage', 'High Usage']
data["Usage_cat"] = pd.cut(data["Usage"], bins = use_bins, labels = use_lbl)
```

*>**Insight:**Here are the bins defined for converting ages, education levels, fitness ratings, incomes, and miles into categories for better analysis:*

Age Categories:

Teen: Ages 0-21

Adult: Ages 21-35

Middle Aged: Ages 35-45

Elder: Ages 45 and above

Education Categories:

Low Edu: Education levels up to 13 years

Moderate Edu: Education levels between 13 and 18 years

High Edu: Education levels above 18 years

Fitness Categories:

Poor Shape: Fitness rating of 1

Bad Shape: Fitness rating of 2

Average Shape: Fitness rating of 3

Good Shape: Fitness rating of 4

Excellent Shape: Fitness rating of 5

Income Categories:

Low: Income up to $40,000

Moderate: Income between $40, 000 and$60,000

High Income: Income between $60, 000 and$80,000

Very High: Income above $80,000

Miles Categories:

Light Activity: Miles up to 50

Moderate Activity: Miles between 50 and 100

Active Lifestyle: Miles between 100 and 200

Fitness Enthusiast: Miles above 200

Usage Categories:

Low Usage: Usage up to 3 times a week

Moderate Usage: Usage between 3 and 5 times a week

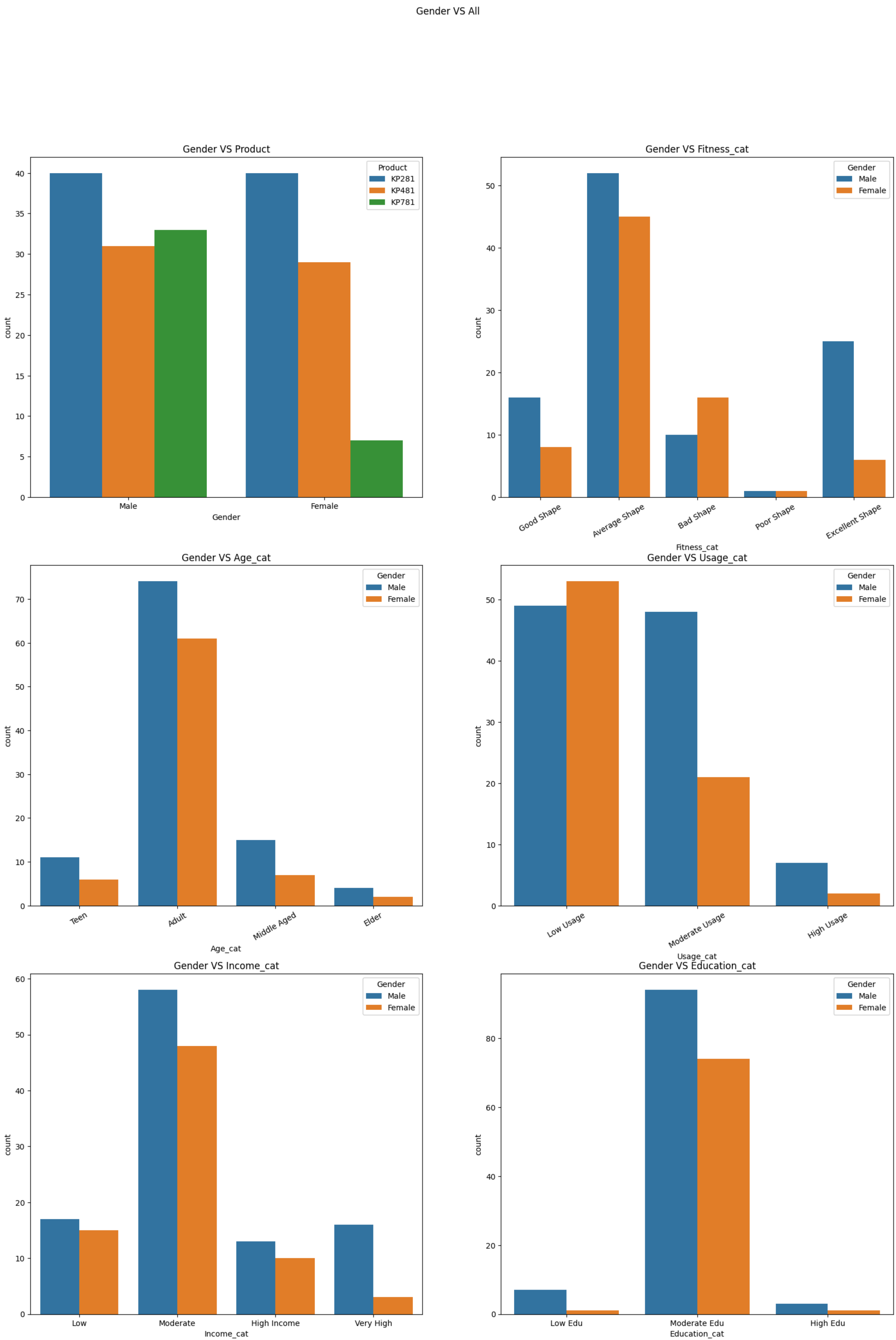
High Usage: Usage more than 5 times a week

These categories will allow for a more structured analysis and insights into the various customer segments based on their demographic and behavioral attributes.

Bivariate Analysis

Gender VS All

```
plt.figure(figsize= (20, 27))
plt.suptitle('Gender VS All')

plt.subplot(3, 2, 1)
sns.countplot(data= data, x= 'Gender', hue= 'Product')
plt.title('Gender VS Product')

plt.subplot(3, 2, 2)
sns.countplot(data= data, hue= 'Gender', x= 'Fitness_cat')
plt.xticks(rotation= 30)
plt.title('Gender VS Fitness_cat')

plt.subplot(3, 2, 3)
sns.countplot(data= data, hue= 'Gender', x= 'Age_cat')
plt.xticks(rotation= 30)
plt.title('Gender VS Age_cat')

plt.subplot(3, 2, 4)
sns.countplot(data= data, hue= 'Gender', x= 'Usage_cat')
plt.xticks(rotation= 30)
plt.title('Gender VS Usage_cat')

plt.subplot(3, 2, 5)
sns.countplot(data= data, hue= 'Gender', x= 'Income_cat')
plt.title('Gender VS Income_cat')

plt.subplot(3, 2, 6)
sns.countplot(data= data, hue= 'Gender', x= 'Education_cat')
plt.title('Gender VS Education_cat')

plt.show()
```



Insight And Recommendation

*>**Insight**:*

Gender vs Product:

Both males and females show a similar preference for KP281 and KP481 treadmills, with usage being evenly distributed between the two genders. However, there is a significant difference in the usage of KP781, with males showing a higher preference compared to females.

Gender vs Fitness Category:

The majority of both males and females fall into the "Average Shape" fitness category. However, there's a noticeable trend where more males tend to be in "Good Shape" or "Excellent Shape" compared to females.

Gender vs Age Category:

Adults show a higher inclination towards fitness activities compared to other age groups. This trend is observed across both genders.

Gender vs Usage Category:

A larger proportion of females tend to fall into the lower usage category, indicating that they use treadmills less frequently compared to males. In contrast, males are more prevalent in the higher and moderate usage categories.

Gender vs Income Category:

The majority of individuals fall into the moderate income category, with a notable gender difference. Males tend to have higher incomes overall, as evidenced by their larger representation in the high and very high-income categories.

Gender vs Education Category:

Most individuals have a moderate level of education, with males being more prevalent in lower education categories. Additionally, there are fewer individuals with higher education levels, with males being more represented in this category as well.

*>**Recommendation:**Focus marketing efforts for KP781 towards females to potentially increase their usage and bridge the gender gap observed in product preference.*

Tailor fitness programs to address the specific needs and preferences of females, considering their tendency towards the "Average Shape" fitness category.

Design marketing campaigns aimed at adults, highlighting the benefits of fitness and health, as this age group shows a higher interest in fitness activities.

Implement strategies to encourage higher treadmill usage among females, such as personalized workout plans or incentives for consistent usage.

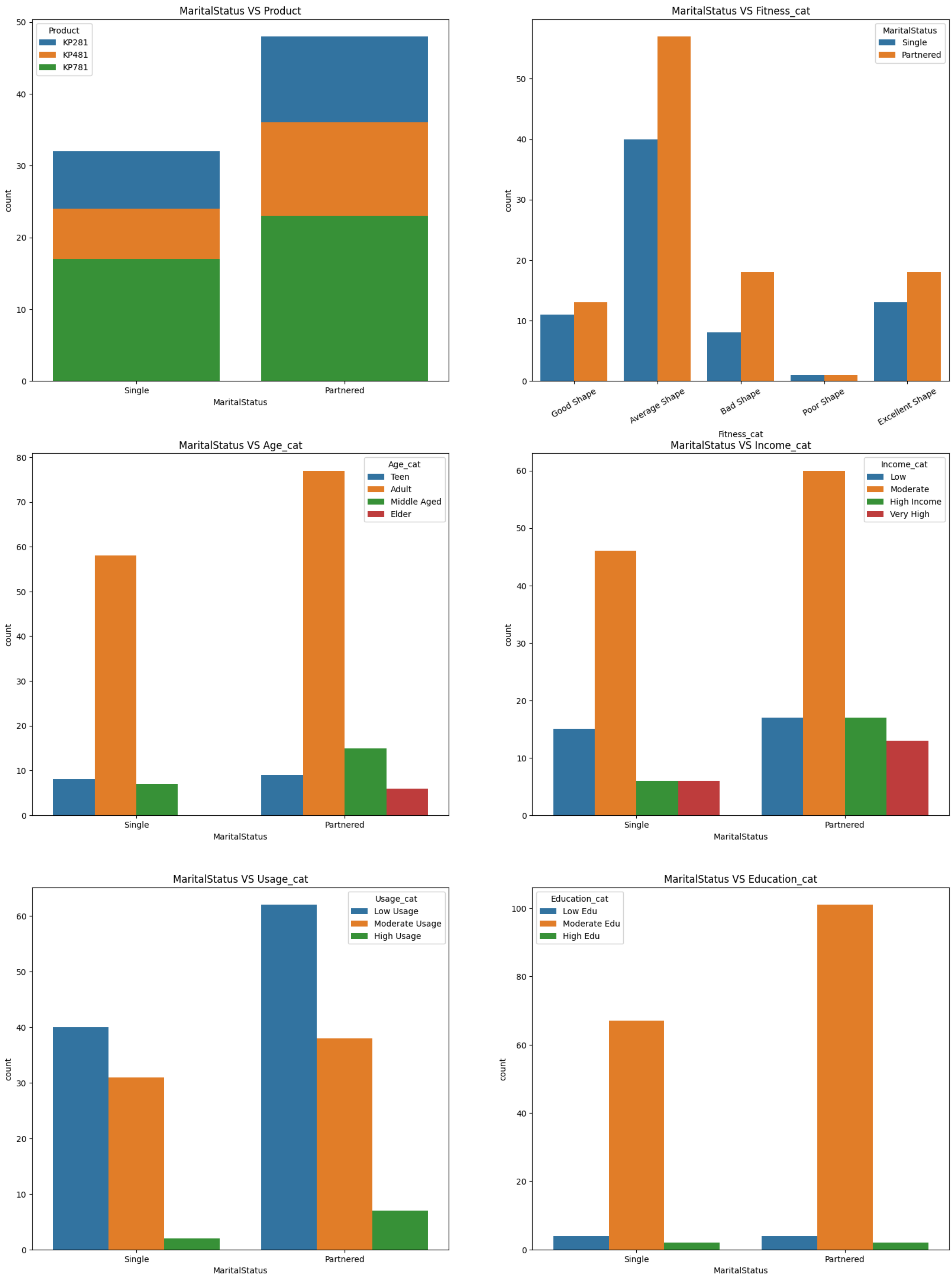
Offer financing options or discounts to make treadmills more accessible to individuals in the moderate income category, considering their prevalence in the dataset.

Develop educational resources or programs aimed at individuals with lower education levels, with a focus on promoting fitness and health awareness, particularly among males.

Marital Status VS All

```
plt.figure(figsize= (20, 27))
plt.suptitle('MaritalStatus VS All')

plt.subplot(3, 2, 1)
sns.countplot(data= data, x= 'MaritalStatus', hue= 'Product', dodge= False)
plt.title('MaritalStatus VS Product')

plt.subplot(3, 2, 2)
sns.countplot(data= data, hue= 'MaritalStatus', x= 'Fitness_cat')
plt.xticks(rotation= 30)
plt.title('MaritalStatus VS Fitness_cat')

plt.subplot(3, 2, 3)
sns.countplot(data= data, x= 'MaritalStatus', hue= 'Age_cat')
plt.title('MaritalStatus VS Age_cat')

plt.subplot(3, 2, 4)
sns.countplot(data= data, x= 'MaritalStatus', hue= 'Income_cat')
plt.title('MaritalStatus VS Income_cat')

plt.subplot(3, 2, 5)
sns.countplot(data= data, x= 'MaritalStatus', hue= 'Usage_cat')
plt.title('MaritalStatus VS Usage_cat')

plt.subplot(3, 2, 6)
sns.countplot(data= data, x= 'MaritalStatus', hue= 'Education_cat')
plt.title('MaritalStatus VS Education_cat')

plt.show()
```

MaritalStatus VS All



Insight And Recommendation

>*Insight*:

Marital Status vs. Product:

Married individuals tend to use fitness products more than singles across all product categories.

Marital Status vs. Fitness Category:

Married individuals exhibit a broader distribution across fitness categories, with higher proportions in "Good," "Average," "Bad," and even "Excellent" shape compared to singles. Conversely, singles are more concentrated in the "Average Shape" category.

Marital Status vs. Age Category:

The majority of adults are married, while there is a smaller proportion of unmarried individuals within the adult age group.

Marital Status vs. Income Category:

Married couples appear to have a tendency towards higher incomes compared to singles. This is indicated by the potential skew towards higher income categories among married individuals.

Marital Status vs. Usage Category:

Married couples likely utilize fitness products more frequently than singles, as suggested by the potential skew towards higher usage categories among married individuals.

Marital Status vs. Education Category:

There is a higher prevalence of married individuals among those with moderate levels of education compared to singles.

*>**Recommendation:**Develop targeted marketing campaigns aimed at married couples, emphasizing the benefits of fitness products tailored to their lifestyle and health needs.*

Offer fitness programs and resources tailored to married couples, addressing their varied fitness levels and preferences.

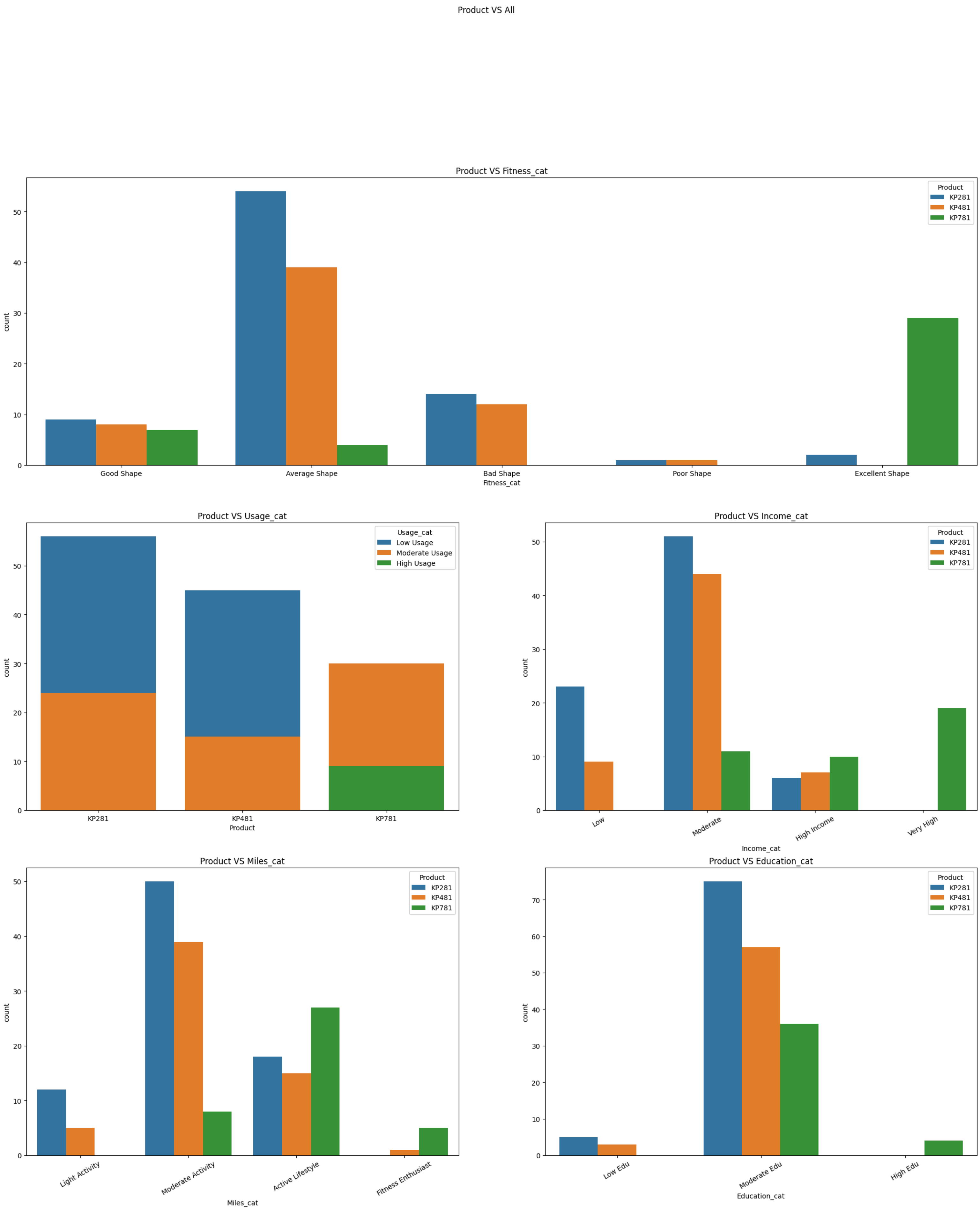
Consider offering financing options or discounts on fitness products for married couples to capitalize on their potentially higher income levels.

Implement strategies to encourage consistent product usage among married couples, such as loyalty programs or personalized workout plans.

Develop educational materials focusing on fitness and health awareness targeted at individuals with moderate education levels, with an emphasis on married couples.

Product VS All

```
plt.figure(figsize= (25, 35))
plt.suptitle('Product VS All')

plt.subplot(4, 1, 1)
sns.countplot(data= data, x='Fitness_cat', hue='Product')
plt.title('Product VS Fitness_cat')

plt.subplot(4, 2, 3)
sns.countplot(data= data, x='Product', hue='Usage_cat', dodge=False)
plt.title('Product VS Usage_cat')

plt.subplot(4, 2, 4)
sns.countplot(data= data, x='Income_cat', hue='Product')
plt.xticks(rotation= 30)
plt.title('Product VS Income_cat')

plt.subplot(4, 2, 5)
sns.countplot(data= data, x='Miles_cat', hue='Product')
plt.xticks(rotation= 30)
plt.title('Product VS Miles_cat')

plt.subplot(4, 2, 6)
sns.countplot(data= data, x='Education_cat', hue='Product')
plt.xticks(rotation= 30)
plt.title('Product VS Education_cat')

plt.show()
```



Insight And Recommendation

*>**Insight**:*

Product vs Fitness Category:

The majority of individuals in "Average Shape" tend to use KP281 and KP481 treadmills, while almost all individuals in "Excellent Shape" prefer KP781.

Product vs Usage Category:

KP281 and KP481 treadmills are associated with moderate and lower usage, while KP781 exhibits a mix of moderate and high usage patterns.

Product vs Income Category:

Lower-income individuals predominantly utilize KP281 and KP481 treadmills, while those with moderate incomes use all three models, albeit with a higher proportion favoring KP281 and KP481. Higher and very high-income individuals show a preference towards KP781.

Product vs Miles Category:

The majority of individuals fall into the "Moderate Activity" category for all three products. However, KP781 sees a higher proportion of users in the "Active Lifestyle" and "Fitness Enthusiast" categories compared to KP281 and KP481.

Product vs Education Category:

Individuals with moderate education levels tend to use all three products, with a higher proportion favoring KP281. KP781 is predominantly used by individuals with higher levels of education.

*>**Recommendation:**Tailor marketing efforts for KP281 and KP481 towards individuals seeking maintenance workouts or beginner-level fitness, while highlighting advanced features and capabilities of KP781 for fitness enthusiasts.*

Implement incentive programs or personalized workout plans to encourage higher usage among KP281 and KP481 users, while emphasizing the premium features and benefits of KP781 to attract more users.

Introduce financing or installment plans for KP781 to make it more accessible to individuals with moderate incomes, thereby expanding its customer base.

Highlight the versatility and suitability of KP781 for users with varying activity levels, emphasizing its capability to cater to both moderate and high-intensity workouts.

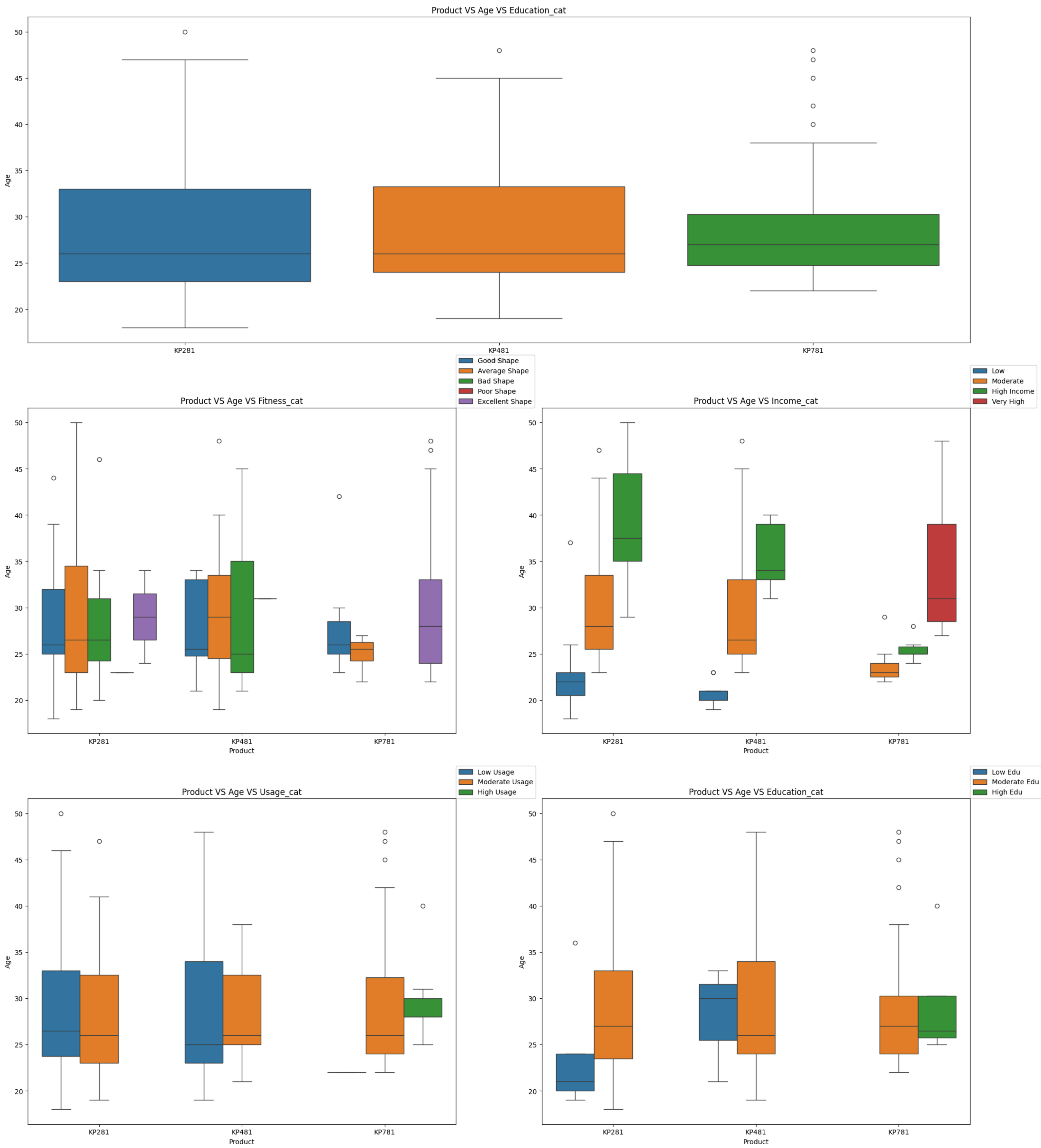
Develop educational content targeting individuals with moderate education levels, focusing on the benefits of using treadmills for overall health and well-being, and highlighting the suitability of different models based on fitness goals.

Multivariate Analysis

Product VS Age VS All

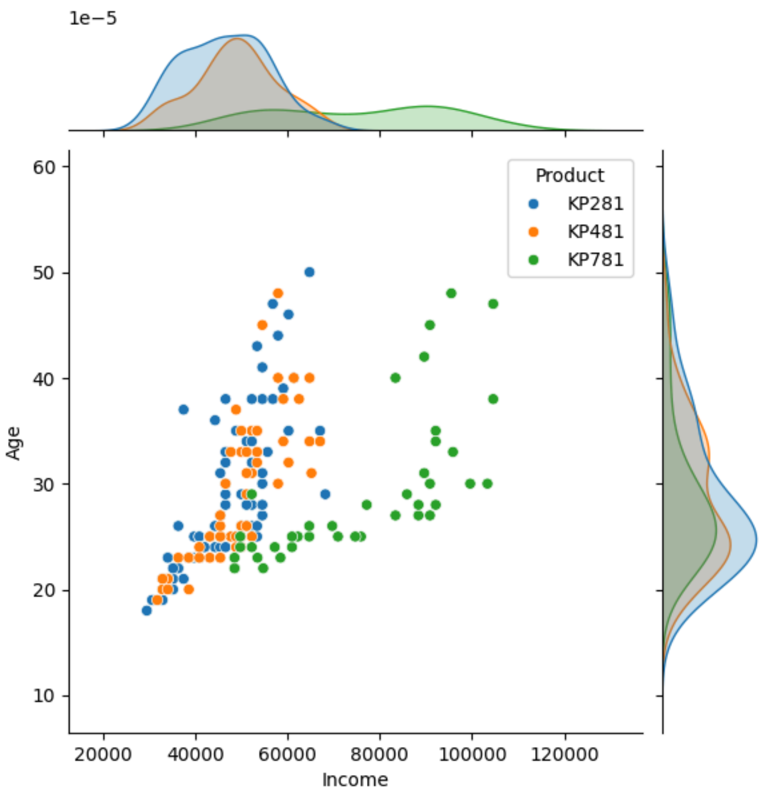
```
plt.figure(figsize= (25, 40))
plt.suptitle('Product VS Age VS All')

plt.subplot(4, 1, 1)
sns.boxplot(data= data, y='Age', x='Product', hue='Product')
plt.title('Product VS Age VS Education_cat')

plt.subplot(4, 2, 3)
sns.boxplot(data= data, y='Age', x='Product', hue='Fitness_cat')
plt.legend(loc= (1,1))
plt.title('Product VS Age VS Fitness_cat')

plt.subplot(4, 2, 4)
sns.boxplot(data= data, y='Age', x='Product', hue='Income_cat')
plt.legend(loc= (1,1))
plt.title('Product VS Age VS Income_cat')

plt.subplot(4, 2, 5)
sns.boxplot(data= data, y='Age', x='Product', hue='Usage_cat')
plt.legend(loc= (1,1))
plt.title('Product VS Age VS Usage_cat')

plt.subplot(4, 2, 6)
sns.boxplot(data= data, y='Age', x='Product', hue='Education_cat')
plt.legend(loc= (1,1))
plt.title('Product VS Age VS Education_cat')

plt.show()
```

Product VS Age VS All



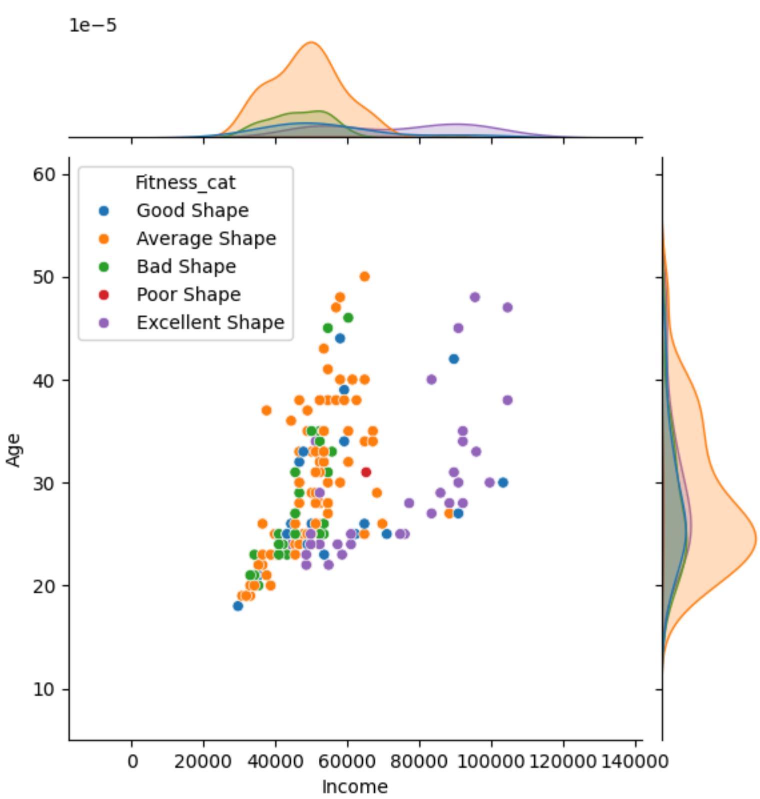
Income VS Age VS Product

```
1 sns.jointplot(data= data, x= 'Income', y= 'Age', hue= 'Product')
2 plt.show()
```



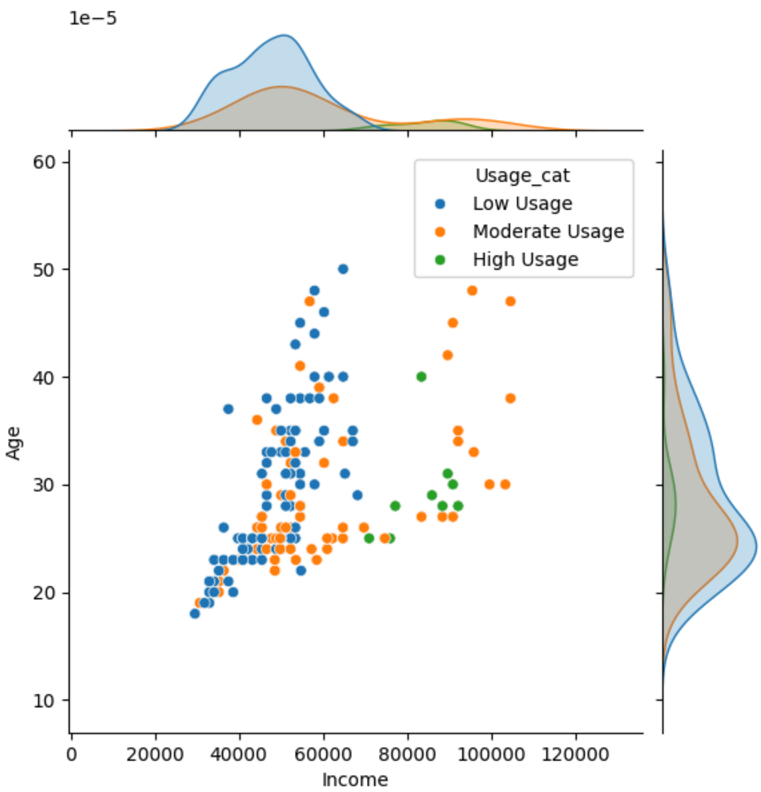
Income VS Age VS Fitness_cat

```
1 sns.jointplot(data= data, x= 'Income', y= 'Age', hue= 'Fitness_cat')
2 plt.show()
```



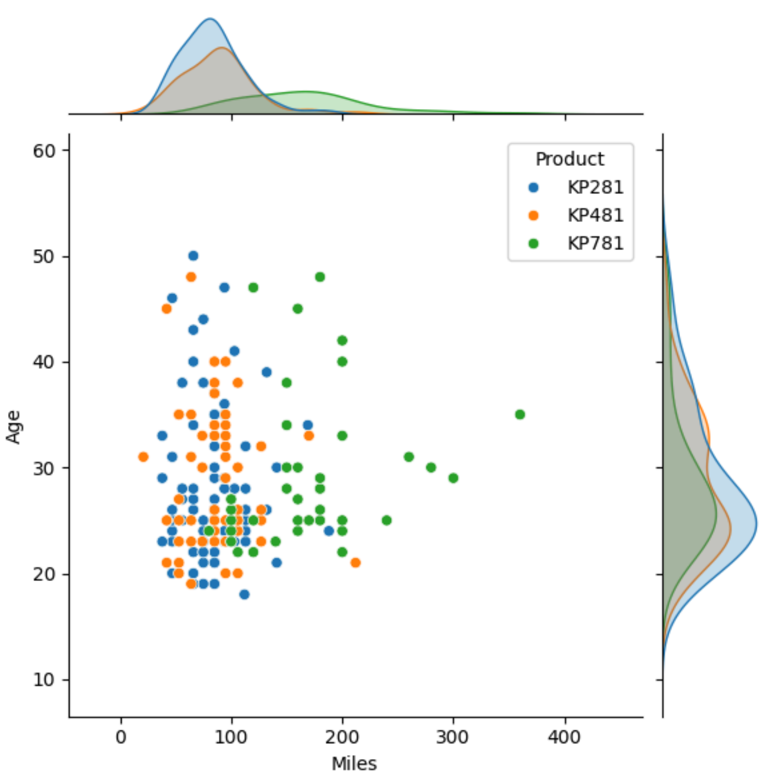
Income VS Age VS Usage_cat

```
1 sns.jointplot(data= data, x= 'Income', y= 'Age', hue= 'Usage_cat')
2 plt.show()
```



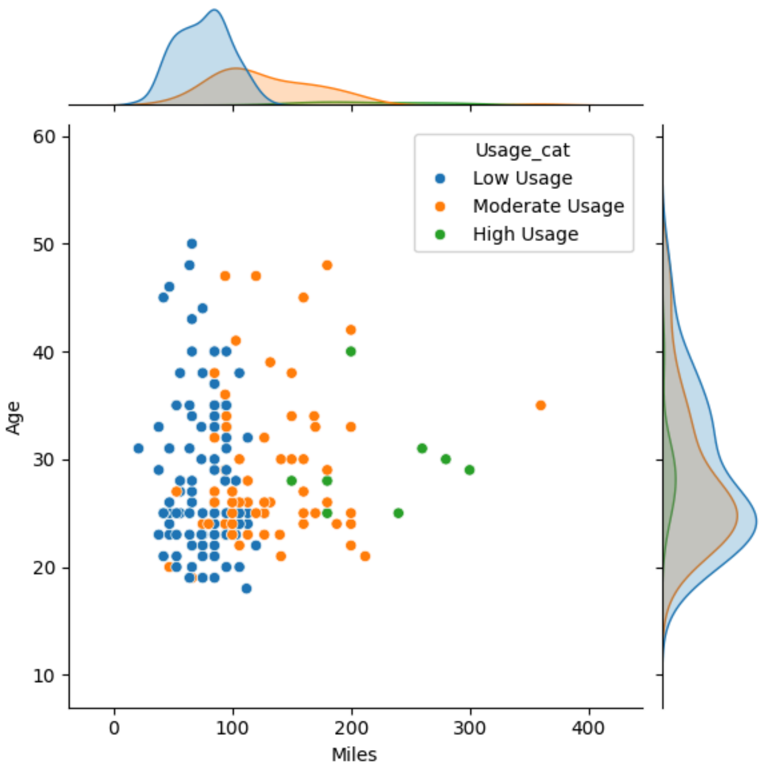
Miles VS Age VS Product

```
1 sns.jointplot(data= data, x= 'Miles', y= 'Age', hue= 'Product')
2 plt.show()
```



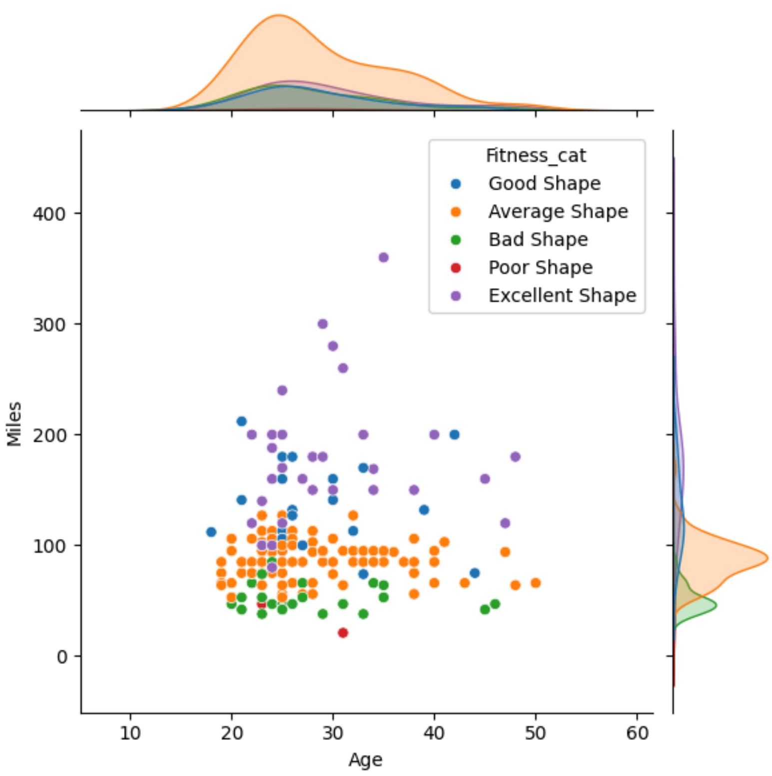
Miles VS Age VS Usage_cat

```
1 sns.jointplot(data= data, x= 'Miles', y= 'Age', hue= 'Usage_cat')
2 plt.show()
```



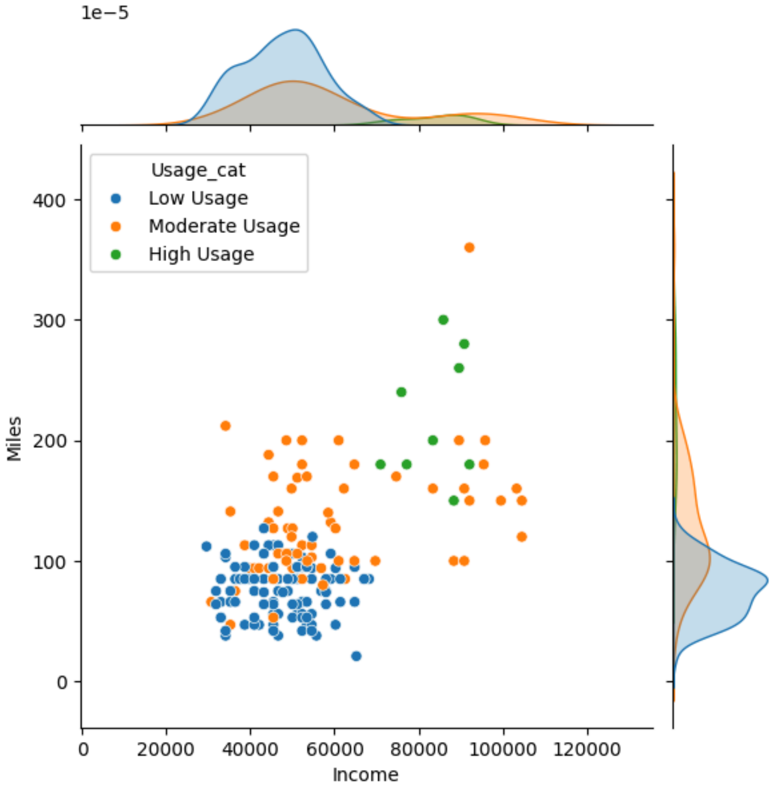
Miles VS Age VS Fitness_cat

```
1 sns.jointplot(data= data, y= 'Miles', x= 'Age', hue= 'Fitness_cat')
2 plt.show()
```



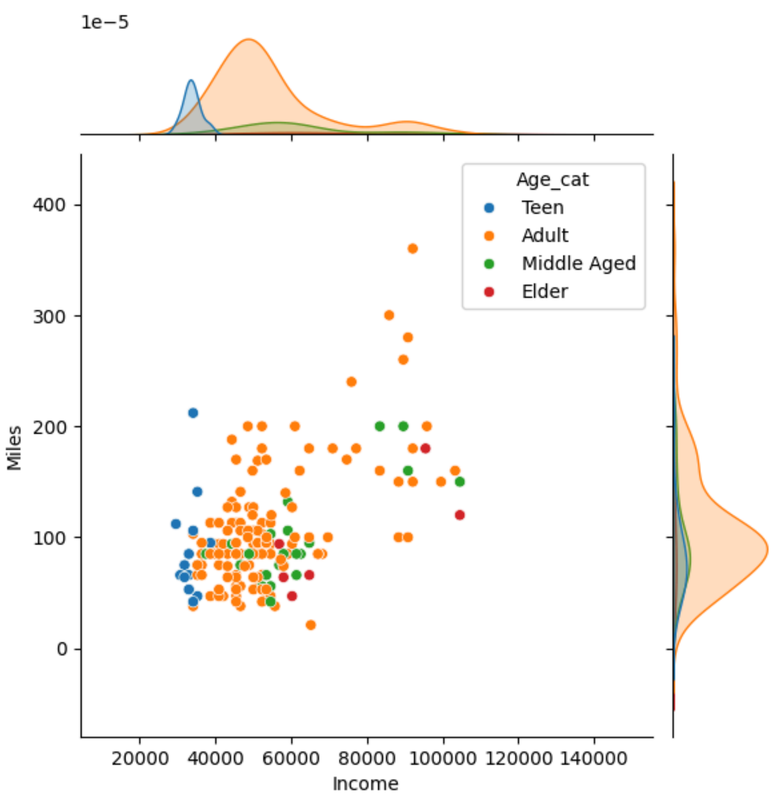
Income VS Miles VS Usage_cat

```
1 sns.jointplot(data= data, x= 'Income', y= 'Miles', hue= 'Usage_cat')
2 plt.show()
```



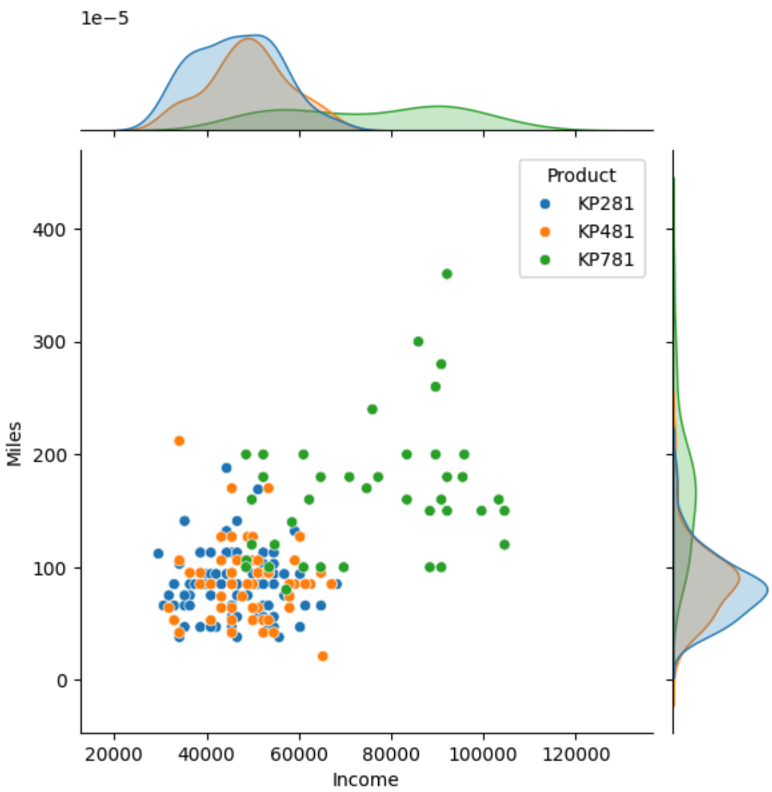
Income VS Miles VS Age_cat

```
1 sns.jointplot(data= data, x= 'Income', y= 'Miles', hue= 'Age_cat')
2 plt.show()
```



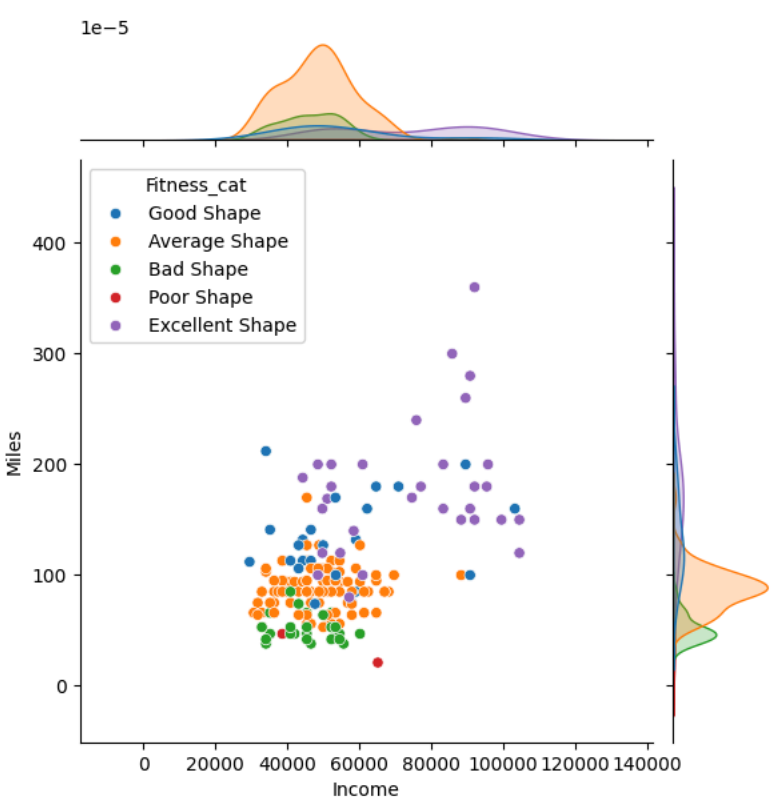
Income VS Miles VS Product

```
1 sns.jointplot(data= data, x= 'Income', y= 'Miles', hue= 'Product')
2 plt.show()
```



Income VS Miles VS Fitness_cat

```
sns.jointplot(data= data, x= 'Income', y= 'Miles', hue= 'Fitness_cat')
plt.show()
```



4. Representing The Probability

a. Find the marginal probability (what percent of customers have purchased KP281, KP481, or KP781)

Hint: We want you to use the pandas crosstab to find the marginal probability of each product.

b. Find the probability that the customer buys a product based on each column.

Hint: Based on previous crosstab values you find the probability.

c. Find the conditional probability that an event occurs given that another event has occurred. (Example: given that a customer is female, what is the probability she'll purchase a KP481)

Hint: Based on previous crosstab values you find the probability.

Marginal And Joint Probability

```
col_cat=['Age_cat', 'Gender', 'Education_cat', 'MaritalStatus']
for col in col_cat:
   print(f'Joint And Marginal Probabily For Product W.R.T {col}:')
   print(pd.crosstab(index= data[col], columns= data['Product'], margins= True, normalize= True) * 100)
   print('===============================================================================================\n\n')
```

```
Joint And Marginal Probabily For Product W.R.T Age_cat:
Product           KP281      KP481      KP781         All
Age_cat
Teen           5.555556   3.888889   0.000000    9.444444
```

```
Adult        31.111111  25.000000  18.888889   75.000000
Middle Aged   6.111111   3.888889   2.222222   12.222222
Elder         1.666667   0.555556   1.111111    3.333333
All          44.444444  33.333333  22.222222  100.000000
==============================================================================================


Joint And Marginal Probabily For Product W.R.T Gender:
Product      KP281      KP481      KP781         All
Gender
Female   22.222222  16.111111   3.888889   42.222222
Male     22.222222  17.222222  18.333333   57.777778
All      44.444444  33.333333  22.222222  100.000000
==============================================================================================


Joint And Marginal Probabily For Product W.R.T Education_cat:
Product             KP281      KP481      KP781         All
Education_cat
Low Edu          2.777778   1.666667   0.000000    4.444444
Moderate Edu    41.666667  31.666667  20.000000   93.333333
High Edu         0.000000   0.000000   2.222222    2.222222
All             44.444444  33.333333  22.222222  100.000000
==============================================================================================


Joint And Marginal Probabily For Product W.R.T MaritalStatus:
Product             KP281      KP481      KP781         All
MaritalStatus
Partnered       26.666667  20.000000  12.777778   59.444444
Single          17.777778  13.333333   9.444444   40.555556
All             44.444444  33.333333  22.222222  100.000000
==============================================================================================
```

```python
col_cat=['Usage_cat', 'Fitness_cat', 'Income_cat', 'Miles_cat']
for col in col_cat:
    print(f'Joint And Marginal Probabily For Product W.R.T {col}:')
    print(pd.crosstab(index= data[col], columns= data['Product'], margins= True, normalize= True) * 100)
    print('==============================================================================================\n\n')
```

```
Joint And Marginal Probabily For Product W.R.T Usage_cat:
Product              KP281      KP481      KP781         All
Usage_cat
Low Usage        31.111111  25.000000   0.555556   56.666667
Moderate Usage   13.333333   8.333333  16.666667   38.333333
High Usage        0.000000   0.000000   5.000000    5.000000
All              44.444444  33.333333  22.222222  100.000000
==============================================================================================


Joint And Marginal Probabily For Product W.R.T Fitness_cat:
Product              KP281      KP481      KP781         All
Fitness_cat
Average Shape    30.000000  21.666667   2.222222   53.888889
Bad Shape         7.777778   6.666667   0.000000   14.444444
Excellent Shape   1.111111   0.000000  16.111111   17.222222
Good Shape        5.000000   4.444444   3.888889   13.333333
Poor Shape        0.555556   0.555556   0.000000    1.111111
All              44.444444  33.333333  22.222222  100.000000
==============================================================================================


Joint And Marginal Probabily For Product W.R.T Income_cat:
Product          KP281      KP481      KP781         All
Income_cat
Low          12.777778   5.000000   0.000000   17.777778
Moderate     28.333333  24.444444   6.111111   58.888889
High Income   3.333333   3.888889   5.555556   12.777778
Very High     0.000000   0.000000  10.555556   10.555556
All          44.444444  33.333333  22.222222  100.000000
==============================================================================================


Joint And Marginal Probabily For Product W.R.T Miles_cat:
Product                  KP281      KP481      KP781         All
Miles_cat
Light Activity        6.666667   2.777778   0.000000    9.444444
Moderate Activity    27.777778  21.666667   4.444444   53.888889
Active Lifestyle     10.000000   8.333333  15.000000   33.333333
Fitness Enthusiast    0.000000   0.555556   2.777778    3.333333
All                  44.444444  33.333333  22.222222  100.000000
==============================================================================================
```

Insight And Recommendation

Product vs. Age Category:

*>**Insight:**Most purchases of all treadmill products (KP281, KP481, KP781) occur within the "Adult" age category, constituting 75% of all purchases. There is a noticeable lack of purchases in the "Elder" age category, accounting for only 3.33% of total purchases.*

*>**Recommendation:**Target marketing efforts towards adults, as they constitute the majority of treadmill purchasers.*

Product vs. Gender:

*>**Insight:**Both genders contribute almost equally to the purchase of KP281 and KP481, with each contributing 22.22% and 16.11%, respectively, to the total purchases. More males purchase KP781, contributing to 18.33% of total purchases compared to 3.88% by females.*

*>**Recommendation:**Tailor advertising messages to resonate with the interests and preferences of both genders, while highlighting features that appeal specifically to males for KP781.*

Product vs. Education Category:

*>**Insight:**The majority of purchases across all treadmill products are made by individuals with a "Moderate Education" level, accounting for 93.33% of total purchases. There's a lack of purchases from individuals with "Low Education" across all products, contributing only 4.44% to the total*

purchases.

*>**Recommendation:**Focus marketing efforts on individuals with a moderate level of education, while considering strategies to attract those with lower education levels.*

Product vs. Marital Status:

*>**Insight:**Partnered individuals tend to make more treadmill purchases compared to singles across all products, constituting 59.44% of total purchases. Singles contribute to 40.56% of total purchases, showing a relatively smaller share.*

*>**Recommendation:**Develop marketing campaigns that emphasize the benefits of treadmill usage for couples or families, while also targeting singles with messaging that promotes individual wellness and fitness goals.*

Product vs. Usage Category:

*>**Insight:**Majority of purchases for all products come from individuals in the "Low Usage" category, contributing to 56.67% of total purchases. There's a significant portion of KP781 purchases from individuals in the "High Usage" category, accounting for 5% of total purchases.*

*>**Recommendation:**Encourage increased usage among customers by promoting the various health benefits associated with consistent treadmill use. Highlight the unique features of KP781 to attract high-usage customers.*

Product vs. Fitness Category:

*>**Insight:**Most purchases are made by individuals in "Average Shape" across all products, contributing to 53.89% of total purchases. KP781 attracts more purchases from individuals in "Excellent Shape" compared to KP281 and KP481, with 16.11% of total purchases.*

*>**Recommendation:**Create targeted marketing materials that emphasize how each product can contribute to maintaining or improving fitness levels, catering to different fitness levels and goals.*

Product vs. Income Category:

*>**Insight:**Majority of purchases are made by individuals with "Moderate Income" across all products, contributing to 58.89% of total purchases. There's a noticeable increase in KP781 purchases from individuals with "High Income" and "Very High Income," accounting for 5.56% and 10.56% of total purchases, respectively.*

*>**Recommendation:**Position KP781 as a premium product with advanced features that justify its higher price point. Highlight its value proposition for customers with higher income levels.*

Product vs. Miles Category:

*>**Insight:**Majority of purchases come from individuals in the "Moderate Activity" category across all products, constituting 53.89% of total purchases. KP781 attracts more purchases from individuals in the "Active Lifestyle" and "Fitness Enthusiast" categories compared to KP281 and KP481, with 15% and 2.78% of total purchases, respectively.*

*>**Recommendation:**Showcase the versatility of KP781 for individuals with active lifestyles or those who are fitness enthusiasts. Highlight its durability and performance capabilities for intensive use.*

Conditional Probability

```
col_cat=['Age_cat', 'Gender', 'Education_cat']
for col in col_cat:
    print(f'Conditional Probabily For Product W.R.T Row Total Of {col}:')
    print(pd.crosstab(index= data[col], columns= data['Product'], margins= True, normalize= 'index') * 100)
    print('----------------------------------------------------------------------------------------------------\n')
    print(f'Conditional Probabily For {col} W.R.T Column Total Of Product:')
    print(pd.crosstab(index= data[col], columns= data['Product'], margins= True, normalize= 'columns') * 100)
    print('====================================================================================================\n\n')
```

```
Conditional Probabily For Product W.R.T Row Total Of Age_cat:
Product          KP281      KP481      KP781
Age_cat
Teen         58.823529  41.176471   0.000000
Adult        41.481481  33.333333  25.185185
Middle Aged  50.000000  31.818182  18.181818
Elder        50.000000  16.666667  33.333333
All          44.444444  33.333333  22.222222
----------------------------------------------------------------------------------------------------

Conditional Probabily For Age_cat W.R.T Column Total Of Product:
Product      KP281      KP481  KP781        All
Age_cat
Teen         12.50  11.666667    0.0   9.444444
Adult        70.00  75.000000   85.0  75.000000
Middle Aged  13.75  11.666667   10.0  12.222222
Elder         3.75   1.666667    5.0   3.333333
====================================================================================================


Conditional Probabily For Product W.R.T Row Total Of Gender:
Product      KP281      KP481      KP781
Gender
Female   52.631579  38.157895   9.210526
Male     38.461538  29.807692  31.730769
All      44.444444  33.333333  22.222222
----------------------------------------------------------------------------------------------------

Conditional Probabily For Gender W.R.T Column Total Of Product:
Product  KP281      KP481  KP781        All
Gender
Female    50.0  48.333333   17.5  42.222222
Male      50.0  51.666667   82.5  57.777778
====================================================================================================


Conditional Probabily For Product W.R.T Row Total Of Education_cat:
Product            KP281      KP481       KP781
Education_cat
Low Edu        62.500000  37.500000    0.000000
Moderate Edu   44.642857  33.928571   21.428571
High Edu        0.000000   0.000000  100.000000
All            44.444444  33.333333   22.222222
----------------------------------------------------------------------------------------------------

Conditional Probabily For Education_cat W.R.T Column Total Of Product:
```

```
Product        KP281  KP481  KP781        All
Education_cat
Low Edu         6.25    5.0    0.0   4.444444
Moderate Edu   93.75   95.0   90.0  93.333333
High Edu        0.00    0.0   10.0   2.222222
=====================================================================================================
```

```python
col_cat=['Usage_cat', 'Fitness_cat', 'Income_cat']
for col in col_cat:
    print(f'Conditional Probabily For Product W.R.T Row Total Of {col}:')
    print(pd.crosstab(index= data[col], columns= data['Product'], margins= True, normalize= 'index') * 100)
    print('----------------------------------------------------------------------------------------------------\n')
    print(f'Conditional Probabily For {col} W.R.T Column Total Of Product:')
    print(pd.crosstab(index= data[col], columns= data['Product'], margins= True, normalize= 'columns') * 100)
    print('=====================================================================================================\n\n')
```

```
Usage_cat
Low Usage       54.901961  44.117647    0.980392
Moderate Usage  34.782609  21.739130   43.478261
High Usage       0.000000   0.000000  100.000000
All             44.444444  33.333333   22.222222
----------------------------------------------------------------------------------------------------

Conditional Probabily For Usage_cat W.R.T Column Total Of Product:
Product         KP281  KP481  KP781        All
Usage_cat
Low Usage        70.0   75.0    2.5  56.666667
Moderate Usage   30.0   25.0   75.0  38.333333
High Usage        0.0    0.0   22.5   5.000000
=====================================================================================================


Conditional Probabily For Product W.R.T Row Total Of Fitness_cat:
Product              KP281      KP481      KP781
Fitness_cat
Average Shape    55.670103  40.206186   4.123711
Bad Shape        53.846154  46.153846   0.000000
Excellent Shape   6.451613   0.000000  93.548387
Good Shape       37.500000  33.333333  29.166667
Poor Shape       50.000000  50.000000   0.000000
All              44.444444  33.333333  22.222222
----------------------------------------------------------------------------------------------------

Conditional Probabily For Fitness_cat W.R.T Column Total Of Product:
Product          KP281      KP481  KP781        All
Fitness_cat
Average Shape    67.50  65.000000   10.0  53.888889
Bad Shape        17.50  20.000000    0.0  14.444444
Excellent Shape   2.50   0.000000   72.5  17.222222
Good Shape       11.25  13.333333   17.5  13.333333
Poor Shape        1.25   1.666667    0.0   1.111111
=====================================================================================================


Conditional Probabily For Product W.R.T Row Total Of Income_cat:
Product          KP281      KP481       KP781
Income_cat
Low          71.875000  28.125000    0.000000
Moderate     48.113208  41.509434   10.377358
High Income  26.086957  30.434783   43.478261
Very High     0.000000   0.000000  100.000000
All          44.444444  33.333333   22.222222
----------------------------------------------------------------------------------------------------

Conditional Probabily For Income_cat W.R.T Column Total Of Product:
Product      KP281      KP481  KP781        All
Income_cat
Low          28.75  15.000000    0.0  17.777778
Moderate     63.75  73.333333   27.5  58.888889
High Income   7.50  11.666667   25.0  12.777778
Very High     0.00   0.000000   47.5  10.555556
=====================================================================================================
```

```python
col_cat=['MaritalStatus', 'Miles_cat']
for col in col_cat:
    print(f'Conditional Probabily For Product W.R.T Row Total Of {col}:')
    print(pd.crosstab(index= data[col], columns= data['Product'], margins= True, normalize= 'index') * 100)
    print('----------------------------------------------------------------------------------------------------\n')
    print(f'Conditional Probabily For {col} W.R.T Column Total Of Product:')
    print(pd.crosstab(index= data[col], columns= data['Product'], margins= True, normalize= 'columns') * 100)
    print('=====================================================================================================\n\n')
```

```
Conditional Probabily For Product W.R.T Row Total Of MaritalStatus:
Product            KP281      KP481      KP781
MaritalStatus
Partnered      44.859813  33.644860  21.495327
Single         43.835616  32.876712  23.287671
All            44.444444  33.333333  22.222222
----------------------------------------------------------------------------------------------------

Conditional Probabily For MaritalStatus W.R.T Column Total Of Product:
Product        KP281  KP481  KP781        All
MaritalStatus
Partnered       60.0   60.0   57.5  59.444444
Single          40.0   40.0   42.5  40.555556
=====================================================================================================


Conditional Probabily For Product W.R.T Row Total Of Miles_cat:
Product                 KP281      KP481      KP781
Miles_cat
Light Activity      70.588235  29.411765   0.000000
Moderate Activity   51.546392  40.206186   8.247423
Active Lifestyle    30.000000  25.000000  45.000000
Fitness Enthusiast   0.000000  16.666667  83.333333
All                 44.444444  33.333333  22.222222
```

```
----------------------------------------------------------------------------------------------------

Conditional Probabily For Miles_cat W.R.T Column Total Of Product:
Product             KP281      KP481  KP781        All
Miles_cat
Light Activity       15.0   8.333333    0.0   9.444444
Moderate Activity    62.5  65.000000   20.0  53.888889
Active Lifestyle     22.5  25.000000   67.5  33.333333
Fitness Enthusiast    0.0   1.666667   12.5   3.333333
=====================================================================================================
```

Insight And Recommendation

Product W.R.T Age Category:

Teenagers predominantly prefer KP281 compared to other age groups, constituting 58.82% of its users. Adults show a more balanced distribution among all three products, with KP781 being the highest at 25.19%. Elderly individuals tend to favor KP281 and KP781 almost equally. It's recommended to target marketing campaigns for KP281 towards teenagers, while focusing on KP781 for adults.

Age Category W.R.T Product:

Among teenagers, KP281 has the highest preference, accounting for 12.5% of its users. Adults exhibit a balanced preference among all products, with KP781 having the highest share at 85%. The elderly population shows varied preferences, with KP281 and KP781 being the most favored. Tailoring product features and marketing strategies according to age demographics can enhance product adoption.

Product W.R.T Gender:

Females show a higher preference for KP281 compared to males, accounting for 52.63% of its users. Males exhibit a more balanced preference among all products, with KP781 being slightly favored. Gender-specific marketing campaigns can be developed to target the preferences observed, potentially increasing sales.

Gender W.R.T Product:

KP281 and KP481 have similar preferences among both genders. KP781 is slightly more preferred by males compared to females. Tailoring product features and marketing messages to suit the preferences of each gender can improve customer engagement.

Product W.R.T Education Category:

Individuals with moderate education levels show the highest preference for all products. Those with low education levels primarily prefer KP281, constituting 62.50% of its users. High education individuals exhibit a strong preference for KP781, constituting 100% of its users. Designing educational campaigns targeted towards specific education categories can effectively reach potential customers.

Education Category W.R.T Product:

KP281 is preferred across all education categories, with the highest preference among individuals with moderate education. KP781 shows a significant preference among individuals with high education levels. Crafting marketing materials that resonate with the educational backgrounds of customers can improve product resonance.

Product W.R.T Marital Status:

Partnered individuals exhibit a slightly higher preference for all products compared to singles. Marketing strategies can be tailored to highlight product benefits that resonate with the lifestyle and needs of partnered individuals.

Marital Status W.R.T Product:

Both partnered and single individuals show similar preferences for all products. While the difference in preference is marginal, understanding marital status dynamics can aid in refining marketing strategies.

Product W.R.T Usage Category:

Low usage individuals show a strong preference for KP281, constituting 54.90% of its users. Moderate usage individuals exhibit a balanced preference among all products. High usage individuals predominantly prefer KP781, constituting 100% of its users. Tailoring product features and marketing messages according to usage patterns can improve customer engagement.

Usage Category W.R.T Product:

KP281 is preferred across all usage categories, with the highest preference among low usage individuals. KP781 shows a strong preference among high usage individuals. Aligning product features with usage preferences can enhance customer satisfaction and loyalty.

Product W.R.T Fitness Category:

Individuals in average shape show a strong preference for KP281 and KP481. Those in excellent shape predominantly prefer KP781. Developing targeted fitness-oriented campaigns for KP781 can attract customers seeking advanced features.

Fitness Category W.R.T Product:

KP281 is preferred across all fitness categories, with the highest preference among individuals in average shape. KP781 shows a strong preference among individuals in excellent shape. Customizing product offerings based on fitness preferences can enhance customer engagement and satisfaction.

Product W.R.T Income Category:

Low-income individuals predominantly prefer KP281, constituting 71.88% of its users. Moderate-income individuals exhibit a balanced preference among all products. High-income individuals show a significant preference for KP781, constituting 43.48% of its users. Developing pricing strategies and promotional offers tailored to income levels can attract customers across different segments.

Income Category W.R.T Product:

KP281 is preferred across all income categories, with the highest preference among low-income individuals. KP781 shows a strong preference among high-income individuals. Offering flexible payment options and value-added services can appeal to customers across various income levels.

Product W.R.T Miles Category:

Individuals engaged in light activity prefer KP281, constituting 70.59% of its users. Moderate activity individuals show a balanced preference among all products. Fitness enthusiasts predominantly prefer KP781, constituting 83.33% of its users. Designing product features and marketing campaigns that align with activity levels can enhance customer satisfaction and retention.

Miles Category W.R.T Product:

KP281 is preferred across all miles categories, with the highest preference among individuals engaged in light activity. KP781 shows a strong preference among fitness enthusiasts. Tailoring product features to meet the needs of customers based on their activity levels can improve product adoption and loyalty.

5. Check The Correlation Among Different Factors

a. Find the correlation between the given features in the table.

Hint: We want you can use the heatmap and corr function to find the correlation between the variables

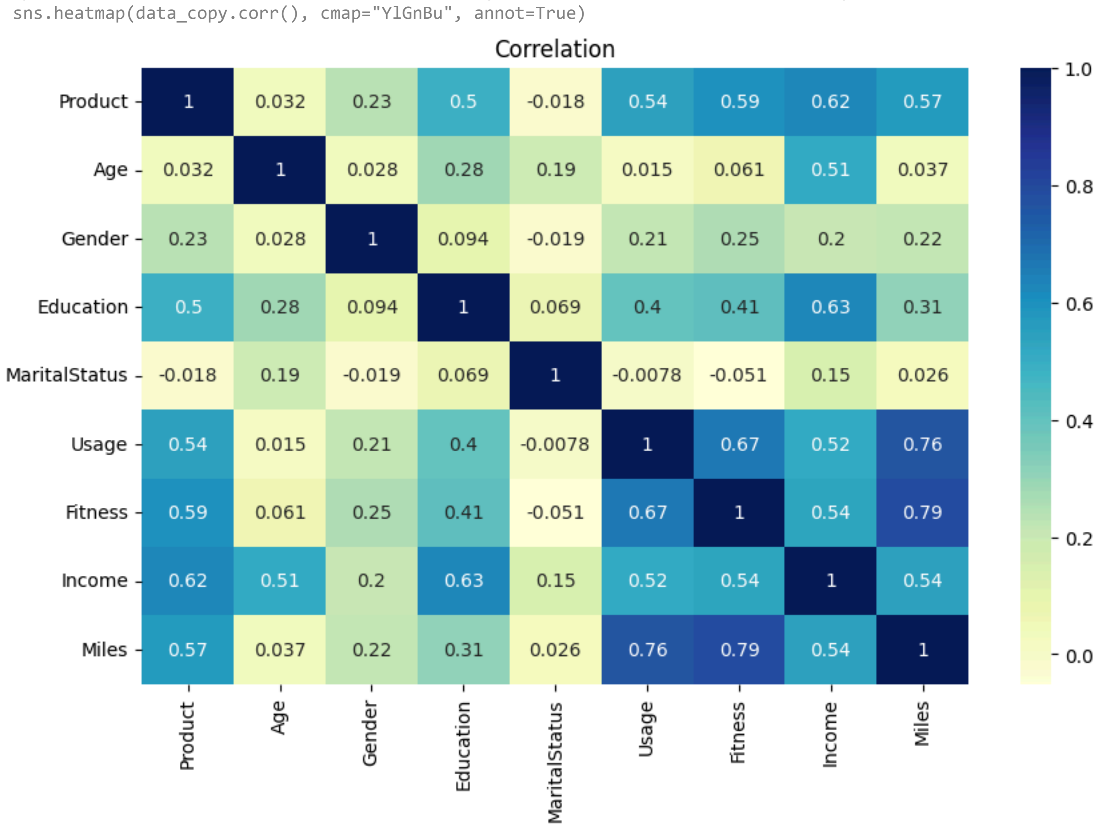
```
#Converting categorical data to numerical datas for better analysis on correlation
data_copy= data.copy()

data_copy['Product'].replace(['KP281', 'KP481', 'KP781'], [0, 1, 2], inplace= True)

data_copy['MaritalStatus'].replace(['Single', 'Partnered'], [0, 1], inplace= True)

data_copy['Gender'].replace(['Female', 'Male'], [0, 1], inplace= True)

data_copy.corr()
```

```
<ipython-input-157-3c5574bdba84>:10: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated. In a future version, it will def
  data_copy.corr()
```

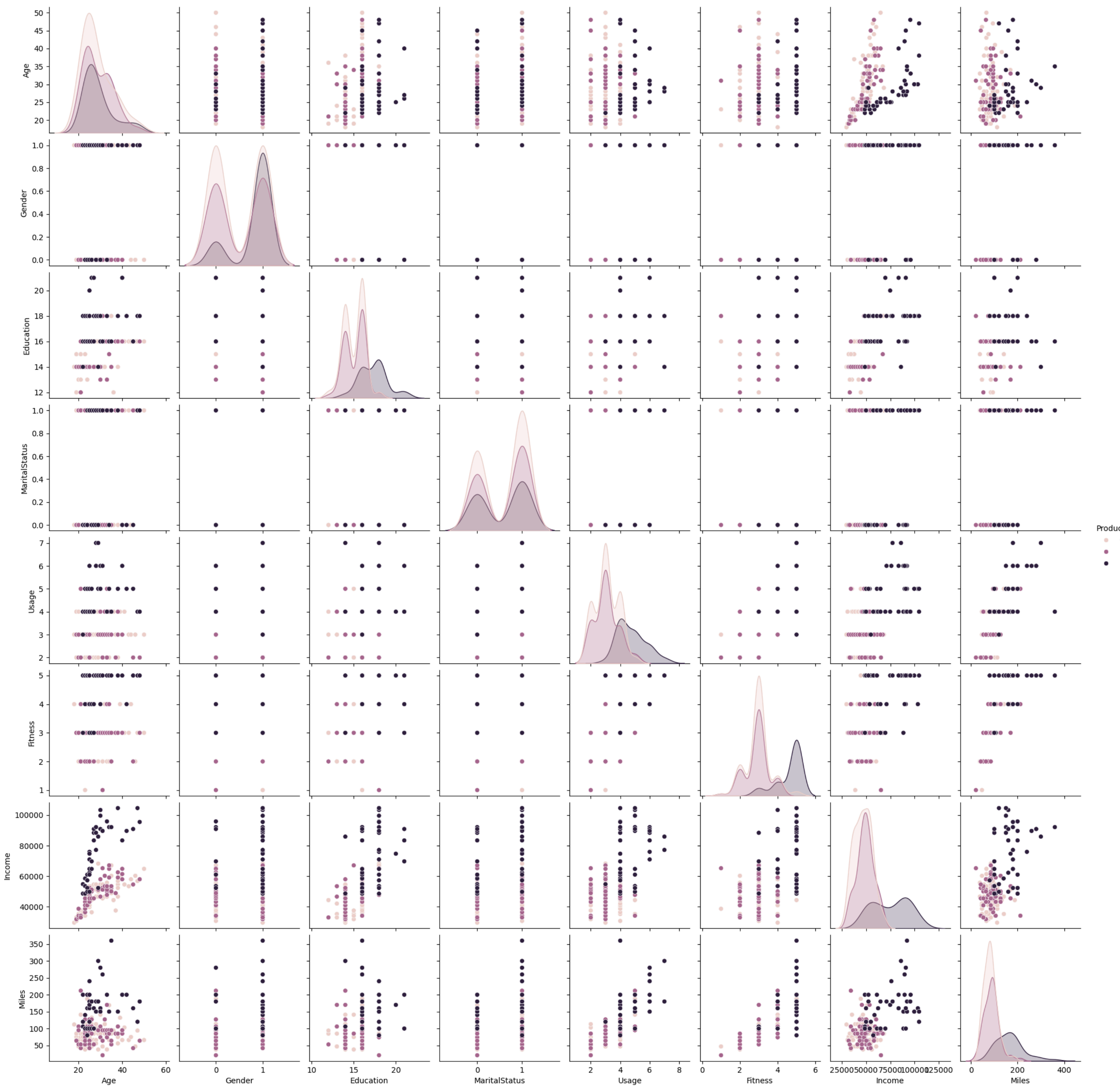
	Product	Age	Gender	Education	MaritalStatus	Usage	Fitness	Income	Miles
Product	1.000000	0.032225	0.230653	0.495018	-0.017602	0.537447	0.594883	0.624168	0.571596
Age	0.032225	1.000000	0.027544	0.280496	0.192152	0.015064	0.061105	0.513414	0.036618
Gender	0.230653	0.027544	1.000000	0.094089	-0.018836	0.214424	0.254609	0.202053	0.217869
Education	0.495018	0.280496	0.094089	1.000000	0.068569	0.395155	0.410581	0.625827	0.307284
MaritalStatus	-0.017602	0.192152	-0.018836	0.068569	1.000000	-0.007786	-0.050751	0.150293	0.025639
Usage	0.537447	0.015064	0.214424	0.395155	-0.007786	1.000000	0.668606	0.519537	0.759130
Fitness	0.594883	0.061105	0.254609	0.410581	-0.050751	0.668606	1.000000	0.535005	0.785702
Income	0.624168	0.513414	0.202053	0.625827	0.150293	0.519537	0.535005	1.000000	0.543473
Miles	0.571596	0.036618	0.217869	0.307284	0.025639	0.759130	0.785702	0.543473	1.000000

```
plt.figure(figsize=(10,6))
sns.heatmap(data_copy.corr(), cmap="YlGnBu", annot=True)
plt.title('Correlation')
plt.show()
```

```
<ipython-input-158-543d3833d015>:2: FutureWarning: The default value of numeric_only in DataFrame.corr is deprecated. In a future version, it will defa
  sns.heatmap(data_copy.corr(), cmap="YlGnBu", annot=True)
```




```
sns.pairplot(data= data_copy, hue= 'Product')
plt.show()
```



Insight And Recommendation

>***Insight:****Products purchased are highly correlated with education, income, usage, fitness, and miles. This suggests that customers with higher education, income levels, and fitness goals tend to opt for certain treadmill models more than others.*

Age shows a significant correlation with income (0.51), which aligns with common expectations. Additionally, age is correlated with education and marital status, indicating that older individuals might have higher income levels and more stable relationships.

Gender correlates with usage, fitness, income, and miles. Men tend to exhibit higher usage, better fitness levels, and potentially higher incomes compared to women. Understanding these differences can help tailor marketing strategies to target specific demographics effectively.

Education is strongly correlated with income, as expected. It also shows connections with age and miles. Higher education levels might lead to better income opportunities, affecting purchasing decisions and fitness activities.

Marital status shows some correlation with income and age. This suggests that marital status can influence financial stability, potentially impacting product choices and fitness behaviors.

Usage is highly correlated with fitness and miles, indicating that individuals who use the treadmill more frequently tend to achieve higher fitness levels and cover more distance. Moreover, usage also shows a notable correlation with income, implying that higher-income individuals might invest more time in fitness activities.

Fitness demonstrates a significant correlation with income, suggesting that individuals with higher incomes may prioritize fitness activities and invest in equipment to support their fitness goals.

>***Recommendation:****Tailor marketing efforts based on demographic insights, such as age, gender, education, and marital status. Highlight product features that resonate with specific customer segments, emphasizing fitness benefits and income-related value propositions.*

Consider developing treadmill models that cater to different demographic preferences and fitness goals. For instance, offer variations that appeal to both beginner and advanced fitness enthusiasts, considering factors like usage intensity and fitness tracking features.

Introduce promotional offers or financing options to make higher-end treadmill models more accessible to a broader range of customers, particularly those with moderate incomes but high fitness aspirations.

Implement loyalty programs or community-building initiatives to encourage consistent treadmill usage and foster a sense of belonging among customers. Offer personalized fitness plans or challenges to keep users motivated and engaged over time.

Continuously analyze customer data to identify emerging trends and preferences, allowing for agile adjustments to product offerings, marketing strategies, and customer engagement initiatives.

Provide educational resources and support services to help customers maximize the benefits of treadmill usage, regardless of their fitness level or experience. This could include online tutorials, personalized training programs, or access to certified fitness coaches.

6. Customer profiling and recommendation

a. Make customer profilings for each and every product.

Insight And Recommendation

Across all tables, the last row labeled "All" provides an overview of the overall probabilities for purchasing the three treadmills: KP281 with a probability of 44.44%, KP481 with 33.33%, and KP781 with 22.22%.

For instance, the conditional probability of purchasing KP281 given an education level of 12 is 66.66%. Similarly, the probability of purchasing KP781 given education levels of 18, 20, or 21 is 82.6%, and for education level 20 or 21, it's 100%.

In terms of usage, the probability of purchasing KP281 given a usage level of 2 is 57.57%, while for KP781, given a usage level of 6 or 7, it's 100%.

For fitness level 2, the probability of purchasing KP481 is 46.15%.

Among individuals aged 30-35, the probability of purchasing KP481 is 53.12%.

If the income exceeds $70,000, theprobabilityofpurchasingKP781is100$*45,000 to $50,000, the probability of purchasing KP481 is 44.11%.*

If the miles run are less than 50, the probability of purchasing KP281 is 70.5%. Conversely, if the miles run exceed 150, the probability of purchasing KP781 is 82.1%.

1. Customer Profiles for KP781:

Individuals with incomes exceeding 70k who have logged over 220 miles exclusively utilize KP781.

Recommendations for KP781 Adoption:

Suggest KP781 for individuals meeting any of the following criteria, in addition to the prerequisite of an income exceeding 70k:

a) Education level equal to or greater than 18.

b) Usage frequency of 5 days or more per week.

c) Fitness level rating of 5.

d) Those who have covered over 150 miles, as 80% of them prefer KP781.

Conditions Against Recommending KP781:

It's advised not to recommend KP781 to individuals meeting any of the following criteria:

a) Education level below 14.

b) Fitness level below 3.

c) Age below 20.

d) Income below 45,000.

e) Miles run fewer than 50.

Why Fewer Women Purchase KP781:

The limited number of women with incomes exceeding 70k can explain the lower proportion of them purchasing KP781, primarily due to affordability constraints.

NOTE: *While KP281 and KP481 don't exhibit significant differences in their costs and the characteristics of their users, a few distinctions have been identified. However, these observations should be validated with additional incremental data to confirm their accuracy and reliability.*

2. Customer Profiles for KP281:

Women with incomes below 70k and age over 40.

Customers with incomes in the range of 60k-70k who use the treadmill for 3 days.

Customers with incomes in the range of 45k-50k who use the treadmill for 2 days.

Customers with incomes in the range of 35k-45k who use the treadmill for 4 days.

Customers with incomes in the range of 50k-60k who use the treadmill for 4 days.

Customers with Fitness level 4, aged around 40, and incomes between 50k-60k.

Customers with an Education level of 16, aged over 32, and incomes between 45k-50k.

Customers with an Education level of 16, aged over 45, and incomes between 60k-70k.

Customers aged between 25-30 and 35-40 with incomes in the range of 35k-45k.

Customers aged 40 and above with incomes between 60k-70k.

Women with incomes less than 35k and who run less than 105 miles.

Customers who use the treadmill 5 days a week, with incomes in the range of 35k-45k, and run more than 140 miles.

Customers with Fitness level 5, incomes under 70k, and incomes between 45k-50k.

Customers with an Education level of 15 and incomes less than 35k.

Customers who use the treadmill for 3 days, run less than 70 miles, and are over 40 years old.

Customers who use the treadmill for 2 days and are aged between 25-30.

3. Customer Profiles for KP481:

Women with incomes below 70k and aged between 32-37.

Customers under the age of 25, with incomes in the range of 50k-60k, and running between 100-150 miles.

Customers with Fitness level 4, aged between 25-32, and incomes between 50k-60k.

Customers with an Education level of 16, aged under 22, and incomes between 45k-50k.

Customers with an Education level of 16, aged under 35, and incomes between 60k-70k.

Customers aged between 35-40 with incomes between 60k-70k.

Women with incomes less than 35k and who run more than 105 miles.

Men with incomes between 60k-70k and who run between 100-150 miles.

Customers with Fitness level 4, incomes under 45k-50k, and who run more than 100 miles.

Customers with an Education level of 13 and incomes in the range of 45k-60k.

Customers who use the treadmill for 2 days and are over 40 years old.

THANK YOU

-Aerofit Case Study By: Nitish Vishwanath Hatapaki

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